

# S.-H. Dan Shim | Curriculum Vitae

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## Degrees

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**2001: Ph.D. Geosciences**, Princeton University, Princeton, New Jersey, USA.

Thesis Adviser: Dr. Thomas S. Duffy

**1994: M.S. Geological Sciences**, Seoul National University, Seoul, Korea.

Thesis Adviser: Drs Soo Jin Kim and Jung Ho Ahn

**1992: B.S. Geological Sciences**, Seoul National University, Seoul, Korea.

## Academic Appointments

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**2021–present: Professor**, School of Earth and Space Exploration, Arizona State University, USA.

**2022–2025: Navrotsky Professor of Materials Research**, College of Liberal Arts and Sciences, Arizona State University, USA.

**2012–2021: Associate Professor**, School of Earth and Space Exploration, Arizona State University, USA.

**2015–present: Honors Faculty**, Barrett Honors College, USA.

**2019: Visiting Professor**, Yonsei Frontier Lab program, Yonsei University, Seoul, Korea.

**2014: Visiting Researcher**, Institut de Physique du Globe de Paris (IPGP), Paris, France.

**2008–2011: Associate Professor of Experimental Geophysics**, Massachusetts Institute of Technology, USA.

**2003–2008: Assistant Professor of Experimental Geophysics**, Massachusetts Institute of Technology, USA.

**2001–2003: Miller Research Fellow**, University of California at Berkeley, USA.

## Research Interests

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Mineralogy of Earth, solar-system planets, and exoplanets. Structure and dynamics of the deep planetary interiors. Magma ocean and early evolution of planets. Interaction between atmosphere and interior. Hydrogen storage and transport in the deep interiors of planets. Structure and dynamics of exoplanets (super-Earths, sub-Neptunes, gas giants, and waterworlds). Materials behavior during planet formation. In-situ measurements of crystal structures, chemical reactions, equations of states, and physical properties in diamond-anvil cell using synchrotron X-ray techniques. Application of X-ray free electron laser techniques for high-pressure research. Development of new materials at high pressures and high temperatures. Hydrogen storage in materials with naturally abundant elements.

## Honors

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**2026:** Guggenheim Fellow in Astronomy and Astrophysics

**2023:** Fellow of Mineralogical Society of America

**2022–2025:** Navrotsky Professorship, Arizona State University

**2017:** Nomination for the College of Liberal Arts and Sciences Zebulon Pearce Distinguished Teaching Award, Arizona State University

**2010:** Doornbos Memorial Prize, Studies of the Earth's Deep Interior, International Union of Geodesy and Geophysics (IUGG)

**2004:** Jephtha H. and Emily V. Wade Award, Massachusetts Institute of Technology

**2001–2003:** Miller Research Fellowship, University of California, Berkeley

**2001:** Graduate Research Award, Mineral and Rock Physics Section, American Geophysical Union (AGU)

**2000:** Outstanding Student Paper Award, Tectonophysics Section, American Geophysical Union (AGU)

**2000–2001:** Charlotte Elizabeth Procter Fellowship, Princeton University

**1999–2000:** Research Board Tuition Award, Princeton University

**1999:** Travel Grant, Association of Princeton Graduate Alumni, Princeton University

**1996:** Hess Fellowship, Department of Geosciences, Princeton University

**1992:** Top-Honors Graduate, College of Natural Sciences, Seoul National University

## Professional Network

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**GoogleScholar:** Shim's records on publication and citations [↗](#)

**ORCID:** Shim's research digital identifier, 0000-0001-5203-6038 [↗](#)

**GitHub:** Shim's open source code and software repository [↗](#)

**ResearchGate:** Shim's publication list with pre-, re-, and post-prints [↗](#)

## Membership

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American Association for the Advancement of Science

American Astronomical Society

American Geophysical Union

Mineralogical Society of America

## Recent Research Collaborations (2022–2025)

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**ASU:** E. Garnero and M. Li (structure of the mantle), K. Leinenweber, T. Sharp, D. Smith, A. Navrotsky, and K. Tsuno (multi-anvil press), P. Buseck (electron microscopy), J. O'Rourke (Mars interior; Earth's core)

**US:** E. Alp (Advanced Photon Source), M. Kunz (Advanced Light Source), W. Mao, A. Gleason, R. Alonso-Mori (Stanford University), V. B. Prakapenka (GSECARS, University of Chicago), D. Stegman (University of California, San Diego), S. Yang (Yale University), S. Tracy, P. Driscoll (Carnegie Institution), I. Oleynik (University of South Florida), S. Hansen (University of Alabama)

**International:** Y. J. Lee (Yonsei University, Korea), G. Morard (Université Grenoble Alpes, France), A. Ravasio (Sorbonne Université, France), S. Speziale (GeoForschungsZentrum Potsdam, Germany), A. Vazan (Open University of Israel)

## Technical Skills

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Laser-heated diamond-anvil cell and multi-anvil press

X-ray diffraction and crystal structure refinement

Dispersive- and gated-Raman spectroscopy

Transmission electron microscopy

Synchrotron X-ray diffraction and spectroscopy: Advanced Photon Source (Argonne National Laboratory), Advanced Light Source (Lawrence Berkeley National Laboratory), National Synchrotron Light Source (Brookhaven National Laboratory), Cornell High Energy Synchrotron Source (Cornell University), and Stanford Synchrotron Radiation Laboratory (Stanford University)

X-ray Free Electron Laser: Linac Coherent Light Source (LCLS) at Stanford Linear Accelerator Center (SLAC) National Accelerator Laboratory, and Pohang Accelerator Laboratory X-ray Free Electron Laser (PAL-XFEL)

Computer programming with Python, FORTRAN, IDL, and C languages and Jupyter in UNIX, Linux, Windows, and Mac OS X systems

# Publication

**Authorship:** The authors from the Shim group are in bold with the following notations:

PI: Shim as the principal investigator

cPI: Shim as a co-principal investigator

UG: Undergraduate students

GS: Graduate students

vGS: Visiting graduate students

PD: Postdocs

**Tags:** Shim's recent developments are highlighted with the following symbols.



Solar-system planets and exoplanets



Technical developments



Educational developments

**Article links:** Click the DOI to open electronic version of an article.

## Published Articles

102. **X. Wei**<sup>GS</sup>, **T. Kim**<sup>PD</sup>, **S. Chen**<sup>PD</sup>, K. Tsuno, M. Kunz, K. Armstrong, S. Chariton, V. B. Prakapenka, and **S.-H. Shim**<sup>PI</sup>. Effect of Si on Mg alloying in metallic iron at high pressures. *Journal of Geophysical Research: Solid Earth*, DOI: 10.1029/2026JB034548 , 2026.  
101. **X. Wei**<sup>GS</sup>, S. Chariton, V. B. Prakapenka, Y. Fei, and **S.-H. Shim**<sup>PI</sup>. Effects of hydrogen on Fe–S alloys and their implications for the Martian core. *Journal of Geophysical Research: Planets*, DOI: 10.1029/2025JE009217 , 2026. 
100. Y. Zhang, K. Bali, C. Dorn, A. Ravasio, H. Yang, S. Pandolfi, A. Chen, **X. Wei**<sup>GS</sup>, L. Libon, Q. Che, D. Zheng, E. Boulard, A. Benuzzi-Mounaix, H. J. Lee, E. Galtier, N. A. Czapla, D. Sokaras, R. Alonso-Mori, A. Gleason, **S.-H. Shim**<sup>cPI</sup>, G. Morard, and W. L. Mao. Shock compression of FeOOH and implications for iron-water interactions in super-Earth magma oceans. *Nature Communications*, DOI: 10.1038/s41467-025-67845-8 , 2025. 
99. W.-Y. Zhou, M. Hao, W. Su, **T. Kim**<sup>PD</sup>, **S. Chen**<sup>PD</sup>, **S.-H. Shim**, D. Zhang, P. Q. H. Nguyen, K. Armstrong, and J. S. Zhang. Elasticity of davemaoite as a primary contributor to lower-mantle heterogeneities. *Science*, DOI: 10.1126/science.adx8356 , 2025.
98. **H. Horn**<sup>GS</sup>, A. Vazan, S. Chariton, V. Prakapenka, and **S.-H. Shim**<sup>PI</sup>. Building wet planets via high-pressure magma-hydrogen reactions. *Nature*, DOI: 10.1038/s41586-025-09630-7 , 2025. 
97. G. Morard, J.-A. Hernandez, C. Pege, C. Nagy, L. Libon, A. Lacquement, D. Sokaras, H. J. Lee, E. Galtier, P. Heimann, E. Cunningham, S. Glenzer, T. Vinci, C. Prescher, S. Boccato, J. Chantel, S. Merkel, Y. Zhang, H. Yang, **X. Wei**<sup>GS</sup>, S. Pandolfi, W. L. Mao, A. E. Gleason, **S.-H. Shim**, R. Alonso-Mori, and A. Ravasio. Structural evolution of liquid silicates under conditions in super-Earth interiors. *Nature Communications*, DOI: 10.1038/s41467-024-51796-7 , 2024.  
96. K. Tsuno, D. S. Grewal, V. Xu, L. Leinbach, K. Leinenweber, A. Wittmann, and **S.-H. Shim**. The effect of nitrogen on the dihedral angle between Fe–Ni melt and ringwoodite: implications for the nitrogen deficit in the bulk silicate Earth. *Geophysical Research Letters*, DOI: 10.1029/2024GL109584 , 2024.
95. K. J. Agboola, S. E. Hansen, E. J. Garnero, S. Rost, M. Li, **S.-H. Shim**. Ultra-low velocity zones beneath the Southern Hemisphere imaged with double-array stacking of *PcP* waveforms. *Journal of Geophysical Research: Solid Earth*, DOI: 10.1029/2023JB028170 , 2024.
94. **T. Kim**<sup>PD</sup>, **X. Wei**<sup>GS</sup>, S. Chariton, V. Prakapenka, Y.-J. Ryu, S. Yang, and **S.-H. Shim**<sup>PI</sup>. Stability of hydrides in sub-Neptune exoplanets with thick hydrogen-rich atmospheres. *Proceedings of the*  

National Academy of Sciences, DOI: 10.1073/pnas.2309786120 [↗](#), 2023.

Press release: [ASU News](#) [↗](#)

93. **T. Kim**<sup>VG</sup>, J. Lee, S. Chariton, V. Prakapenka, R. Husband, N. Giordano, H.-P. Liermann, **S.-H. Shim**<sup>PI</sup>, and Y. Lee. Impact of deeply subducted water on the origin of the  $E'$  layer in the core. *Nature Geosciences*, DOI: 10.1038/s41561-023-01324-x [↗](#), 2023.

Press release: [ASU News](#) [↗](#)

92. **S.-H. Shim**<sup>PI</sup>, **B. Ko**<sup>GS</sup>, D. Sokaras, B. Nagler, H. J. Lee, E. Galtier, S. Glenzer, E. Garanados, T. Vinchi, G. Fiquet, **J. Dolinski**<sup>UG</sup>, **J. Tappan**<sup>UG</sup>, **B. Kulka**<sup>UG</sup>, W. L. Mao, G. Morard, A. Ravasio, A. E. Gleason, and R. Alonso-Mori. Ultrafast X-ray detection of low spin iron in molten silicate at deep planetary interior conditions. *Science Advances*, DOI: 10.1126/sciadv.adi6153 [↗](#), 2023.

Press release: [ASU News](#) [↗](#), [Phys.org](#) [↗](#)

91. **S. Fu**<sup>PD</sup>, S. Chariton, V. B. Prakapenka, and **S.-H. Shim**<sup>PI</sup>. Hydrogen and silicon effects on hexagonal close packed Fe alloys at high Pressure: Implications for the composition of Earth's inner core. *Journal of Geophysical Research*, DOI: 10.1029/2022JB026016 [↗](#), 2023.

90. S. E. Hansen, E. J. Garnero, M. Li, **S.-H. Shim**, and S. Rost. Globally distributed subducted materials along the Earth's core-mantle boundary: Implications for ultralow velocity zones. *Science Advances*, DOI: 10.1126/sciadv.add4838 [↗](#), 2023.

Press release: [University of Alabama News](#) [↗](#), [ASU News](#) [↗](#), [Phys.org](#) [↗](#), [BBC News](#) [↗](#), [NSF Research News](#) [↗](#)

89. **H. Piet**<sup>PD</sup>, A. V. G. Chizmeshya, B. Chen, S. Chariton, E. Greenberg, V. B. Prakapenka, P. R. Buseck, **S.-H. Shim**<sup>PI</sup>. Superstoichiometric alloying of H and close-packed FeNi under high pressures: Implications for hydrogen storage in planetary cores. *Geophysical Research Letters*, DOI: 10.1029/2022GL101155 [↗](#), 2023.

88. **S. Fu**<sup>PD</sup>, S. Chariton, V. B. Prakapenka, and **S.-H. Shim**<sup>PI</sup>. Core origin of seismic velocity anomalies at the Earth's core-mantle boundary. *Nature*, DOI: 10.1038/s41586-023-05713-5 [↗](#), 2023.

Press release: [ASU News](#) [↗](#), [Science Highlights - The Advanced Photon Source](#) [↗](#), [VICE](#) [↗](#), [Argonne National Laboratory Press Release](#) [↗](#)

87. **H. Horn**<sup>GS</sup>, V. Prakapenka, S. Chariton, S. Speziale, and **S.-H. Shim**<sup>PI</sup>. Reaction between hydrogen and ferrous/ferric oxides at high pressures and high temperatures – implications for sub-Neptunes and super-Earths. *Planetary Science Journal*, DOI: 10.3847/PSJ/acab03 [↗](#), 2023.

86. **S. Fu**<sup>PD</sup>, S. Chariton, V. B. Prakapenka, A. Chizmeshya, and **S.-H. Shim**<sup>PI</sup>. Hydrogen solubility in FeSi alloy phases at high pressures and temperatures. *American Mineralogist*, DOI: 10.2138/am-2022-8295 [↗](#), 2022.

85. **B. Ko**<sup>GS</sup>, E. Greenberg, V. Prakapenka, E. Alp, W. Bi, Y. Meng, D. Zhang, and **S.-H. Shim**<sup>PI</sup>. Calcium dissolution in bridgmanite in the Earth's deep mantle. *Nature*, DOI: 10.1038/s41586-022-05237-4 [↗](#), 2022.

Press release: [ASU News](#) [↗](#), [Phys.org](#) [↗](#)

84. **B. Ko**<sup>GS</sup>, S. Chariton, V. Prakapenka, B. Chen, E. J. Garnero, M. Li, and **S.-H. Shim**<sup>PI</sup>. Water-induced diamond formation at Earth's core-mantle boundary. *Geophysical Research Letters*, DOI: 10.1029/2022GL098271 [↗](#), 2022.

Press release: [ASU News](#) [↗](#), [Newsweek](#) [↗](#)

83. **S. Fu**<sup>PD</sup>, S. Chariton, V. B. Prakapenka, A. Chizmeshya, and **S.-H. Shim**<sup>PI</sup>. A new hexagonal ternary alloy phase stable in Fe-Si-H at 28.6–42.2 GPa and 3000 K. *Physical Review B*, DOI: 10.1103/PhysRevB.105.104111 [↗](#), 2022.  
82. **S.-H. Shim**<sup>PI</sup>, A. Chizmeshya, and K. Leinenweber. Water in the crystal structure of CaSiO<sub>3</sub> perovskite. *American Mineralogist*, DOI: 10.2138/am-2022-8009 [↗](#), 2022. 
81. **H. Piet**<sup>PD</sup>, A. V. G. Chizmeshya, B. Chen, S. Chariton, E. Greenberg, V. B. Prakapenka, P. R. Buseck, **S.-H. Shim**<sup>PI</sup>. Effect of nickel on the high-pressure phases in Fe–H. *Physical Review B*, DOI: 10.1103/PhysRevB.104.224106 [↗](#), 2021. 
80. **H. Piet**<sup>PD</sup>, K. Leinenweber, E. Greenberg, V. B. Prakapenka, and **S.-H. Shim**<sup>PI</sup>. Effects of hydrogen on the phase relations in Fe–FeS at pressures of Mars-sized bodies. *Journal of Geophysical Research: Planet*, DOI: 10.1029/2021JE006942 [↗](#), 2021. 
79. **T. Kim**<sup>VG</sup>, S. Chariton, V. Prakapenka, A. Pakhomova, H.-P. Liermann, Z. Liu, S. Speziale, **S.-H. Shim**<sup>CP</sup>, and Y. Lee. Atomic scale mixing between MgO and H<sub>2</sub>O in the deep interiors of water-rich planets. *Nature Astronomy*, DOI: 10.1038/s41550-021-01368-2 [↗](#), 2021.   
*Press release: Salon* [↗](#), *ASU* [↗](#), *Nature Astronomy News and Views* [↗](#).
78. R. J. Husband, R. S. McWilliams, E. J. Pace, A. L. Coleman, H. Hwang, J. Choi, **T. Kim**<sup>VG</sup>, G. Hwang, O. B. Ball, S. H. Chun, D. Nam, S. Kim, H. Cynn, V. B. Prakapenka, **S.-H. Shim**, S. Toleikis, M. I. McMahon, Y. Lee, and H.-P. Liermann. X-ray laser heating of water with gold at high static pressure. *Communications Materials*, DOI: 10.1038/s43246-021-00158-7 [↗](#), 2021.  
77. E. Kohler, C. He, S. E. Moran, **S.-H. Shim**, K. K. Brugman, A. C. Johnson, P. C. Vergeli, M. A. Thompson, H. Graham, M. S. Gudipati, B. Fleury, and B. L. Henderson. The importance of prioritizing exoplanet experimental facilities. *Bulletin of the AAS*, 53(4), DOI: 10.3847/25c2cfef.85cfe526 [↗](#), 2021.  
76. **T. Kim**<sup>VG</sup>, **B. Ko**<sup>GS</sup>, E. Greenberg, V. B. Prakapenka, **S.-H. Shim**<sup>CP</sup>, and Y. Lee. Low melting temperature of anhydrous mantle materials at the core-mantle boundary. *Geophysical Research Letters*, DOI: 10.1029/2020GL089345 [↗](#), 2020.
75. **H. Chen**<sup>GS</sup>, S.-Y. Xie, **B. Ko**<sup>GS</sup>, **T. Kim**<sup>VG</sup>, **C. Nisr**<sup>PD</sup>, V. B. Prakapenka, E. Greenberg, D. Zhang, W. Bi, E. Alp, Y. Lee, and **S.-H. Shim**<sup>PI</sup>. A new iron hydroxide phase stable in hydrous lower-mantle systems. *Earth and Planetary Science Letters*, DOI: 10.1016/j.epsl.2020.116551 [↗](#), 2020. 
74. **H. Horn**<sup>GS</sup>, E. Garhart, K. Leinenweber, V. B. Prakapenka, E. Greenburg, and **S.-H. Shim**<sup>PI</sup>. Oxidation of the interiors of carbide exoplanets. *Planetary Science Journal*, DOI: 10.3847/PSJ/abaa3e [↗](#), 2020.   
*Press release: CNN* [↗](#), *Pre-print is available at arXiv:2005.03175* [↗](#).
73. **C. Nisr**<sup>PD</sup>, **H. Chen**<sup>GS</sup>, K. Leinenweber, A. V. G. Chizmeshya, V. B. Prakapenka, C. Prescher, S. Tkachev, Y. Meng, Z. Liu, and **S.-H. Shim**<sup>PI</sup>. Large H<sub>2</sub>O solubility in dense silica and its implications for the interiors of water-rich planets. *Proceedings of the National Academy of Sciences*, DOI: 10.1073/pnas.1917448117 [↗](#), 2020.    
*Press release: Selected for Media Highlight by PNAS, ASU Now* [↗](#), *Advanced Photon Source Science Highlights* [↗](#), *Phys.org* [↗](#), *Argonne National Laboratory Press Release* [↗](#)
72. J.-A. Hernandez, G. Morard, M. Guarguaglini, R. Alonso-Mori, A. Benuzzi-Mounaix, R. Bolis, G. Fiquet, E. Galtier, A. E. Gleason, S. Glenzer, F. Guyot, **B. Ko**<sup>GS</sup>, H. J. Lee, W. L. Mao, B. Nagler, N. Ozaki, A. K. Schuster, **S.-H. Shim**, T. Vinci, and A. Ravasio. Direct observation of shock-induced amorphization of enstatite. *Geophysical Research Letters*, DOI: 10.1073/pnas.1920470117 [↗](#), 2020. 

71. C. M. Lisse, S. J. Desch, C. T. Unterborn, S. R. Kane, P. R. Young, H. E. Hartnett, N. R. Hinkel, **S.-H. Shim**, E. E. Mamajek, and N. R. Izenberg. A geologically robust procedure for observing rocky exoplanets to maximize the likelihood that atmospheric oxygen is an Earth-Like biosignature. *Astrophysical Journal Letters*, DOI: 10.3847/2041-8213/ab9b91 , 2020.
70. H. Li, Y. Qin, **B. Ko**<sup>GS</sup>, D. B. Trivedi, Y. M. Sayad, L. Liu, **S.-H. Shim**, H. Zhuang, and S. Tongay. Anomalous behavior of 2D Janus monolayers under extreme pressures. *Advanced Materials*, DOI: 10.1002/adma.202002401 , 2020.
69. G. Morard, J.-A. Hernandez, M. Guarguaglini, R. Bolis, A. Benuzzi-Mounaix, T. Vinci, G. Fiquet, M. A. Baron, **S.-H. Shim**<sup>CP</sup>, **B. Ko**<sup>GS</sup>, A. E. Gleason, W. L. Mao, R. Alonso-Mori, H. J. Lee, B. Nagler, E. Galtier, D. Sokaras, S. Glenzer, D. Andrault, G. Garbarino, M. Mezouar, A. Schuster, and A. Ravasio. In situ X-ray diffraction of silicate liquids and glasses under dynamic and static compression to megabar pressures. *Proceedings of the National Academy of Sciences*, DOI: 10.1073/pnas.1920470117 , 2020.
68. H. Hwang, E. Galtier, H. Cynn, I. Eom, S. H. Chun, Y. Bang, G. C. Hwang, J. Choi, **T. Kim**<sup>VGS</sup>, M. Kong, S. Kwon, K. Kang, H. J. Lee, C. Park, J.-I. Lee, Y. Lee, W. Yang, **S.-H. Shim**, T. Vogt, S. Kim, J. Park, S. Kim, D. Nam, J. H. Lee, H. Hyun, M. Kim, T.-Y. Koo, C.-C. Kao, T. Sekine, and Y. Lee. Sub-nanosecond phase transition dynamics in laser-shocked iron. *Science Advances*, DOI: 10.1126/sciadv.aaz5132 , 2020.
67. **H. Piet**<sup>PD</sup>, K. Leinenweber, **J. Tappan**<sup>UG</sup>, E. Greenberg, V. B. Prakapenka, P. R. Buseck, and **S.-H. Shim**<sup>PI</sup>. Dehydration of  $\delta$ -AlOOH in Earth's deep lower mantle: Implications for the deep hydrogen and oxygen cycles. *Minerals*, 10, 384, DOI: 10.3390/min10040384 , 2020.
66. **H. Chen**<sup>GS</sup>, K. Leinenweber, V. B. Prakapenka, M. Kunz, H. Bechtel, Z. Liu, and **S.-H. Shim**<sup>PI</sup>. Phase transformation of hydrous ringwoodite to the lower-mantle phases and the formation of dense hydrous silica. *American Mineralogist*, DOI: 10.2138/am-2020-7261 , 2020.
65. **B. Ko**<sup>GS</sup>, V. B. Prakapenka, M. Kunz, C. Prescher, K. Leinenweber, and **S.-H. Shim**<sup>PI</sup>. Mineralogy and density of Archean volcanic crust in the mantle transition zone. *Physics of the Earth and Planetary Interiors*, DOI: 10.1016/j.pepi.2020.106490 , 2020.
64. L. Bindi, **S.-H. Shim**, T. Sharp, and X. Xie. Evidence for the charge disproportionation of iron in extraterrestrial bridgmanite. *Science Advances*, 6, eaay7893, DOI: 10.1126/sciadv.aay7893 , 2020.  
*Press release: National Science Foundation Research News* , *ASU Now* 
63. **B. L. Kulka**<sup>UG</sup>, **J. Dolinski**<sup>UG</sup>, K. Leinenweber, V. B. Prakapenka, and **S.-H. Shim**<sup>PI</sup>. The bridgmanite–akimotoite–majorite triple point determined in large volume press and laser-heated diamond anvil cell. *Minerals*, 10, 67, DOI: 10.3390/min10010067 , 2020.
62. **H. Chen**<sup>GS</sup>, K. Leinenweber, V. B. Prakapenka, C. Prescher, Y. Meng, H. Bechtel, M. Kunz, and **S.-H. Shim**<sup>PI</sup>. Possible H<sub>2</sub>O storage in the crystal structure of CaSiO<sub>3</sub> perovskite. *Physics of the Earth and Planetary Interiors*, 299, 106412, DOI: 10.1016/j.pepi.2019.106412 , 2019.  
*Press release: Advanced Light Source Science Briefs* 
61. **J. G. O'Rourke**<sup>PD</sup> and **S.-H. Shim**. Hydrogenation of the Martian core by hydrated mantle minerals with implications for the early dynamo. *Journal of Geophysical Research: Planets*, 124, 3422–3441, DOI: 10.1029/2019JE005950 , 2019.

60. **H. Chen**<sup>GS</sup>, S. Zhou, D. Morgan, V. B. Prakapenka, E. Greenberg, K. Leinenweber, and **S.-H. Shim**<sup>PI</sup>. The O–O bonding and hydrogen storage in the pyrite-type PtO<sub>2</sub>. *Inorganic chemistry*, 58, 8300–8307, DOI: 10.1021/acs.inorgchem.9b00046 [↗](#), 2019. 
59. **Y. Ye**<sup>PD</sup>, **S.-H. Shim**<sup>PI</sup>, V. B. Prakapenka, and Y. Meng. Equation of state of solid Ne inter-calibrated with the MgO, Au, Pt, NaCl-B2, and ruby pressure scales up to 130 GPa. *High Pressure Research*, 38, 377–395, DOI: 10.1080/08957959.2018.1493477 [↗](#), 2018.
58. **H. Chen**<sup>GS</sup>, **S.-H. Shim**<sup>PI</sup>, K. Leinenweber, V. B. Prakapenka, Y. Meng, and C. Prescher. Crystal Structure of CaSiO<sub>3</sub> perovskite at 28–62 GPa and 300 K under quasi-hydrostatic stress conditions. *American Mineralogist*, 103, 462–468, DOI: 10.2138/am-2018-6087 [↗](#), 2018.
57. **C. Nisr**<sup>PD</sup>, **S.-H. Shim**<sup>PI</sup>, and K. Leinenweber. Raman spectroscopy of water-rich stishovite and dense high-pressure silica up to 55 GPa. *American Mineralogist*, 102, 2180–2189, DOI: 10.2138/am-2017-5944 [↗](#), 2017.
56. **S.-H. Shim**<sup>PI</sup>, **B. Grocholski**<sup>PD</sup>, **Y. Ye**<sup>PD</sup>, E. Alp, S. Xu, D. Morgan, Y. Meng, and V. B. Prakapenka. Stability of ferrous-iron-rich bridgmanite under reducing mid-mantle conditions. *Proceedings of the National Academy of Sciences*, 114, 6468–6473, DOI: 10.1073/pnas.1614036114 [↗](#), 2017.  
*Press release: Advanced Photon Source Science Highlights* [↗](#)
55. **C. Nisr**<sup>PD</sup>, K. Leinenweber, V. B. Prakapenka, C. Prescher, S. Tkachev, and **S.-H. Shim**<sup>PI</sup>. Phase transition and equation of state of dense hydrous silica up to 63 GPa. *Journal of Geophysical Research: Solid Earth*, 122, 6972–6983, DOI: 10.1002/2017JB014055 [↗](#), 2017.
54. **Y. Ye**<sup>PD</sup>, V. B. Prakapenka, Y. Meng, and **S.-H. Shim**<sup>PI</sup>. Inter-comparison of the gold, platinum, and MgO pressure scales up to 140 GPa and 2,500 K. *Journal of Geophysical Research: Solid Earth*, 122, 3450–3464, DOI: 10.1002/2016JB013811 [↗](#), 2017.
53. **C. Nisr**<sup>PD</sup>, Y. Meng, A. MacDowell, J. Yan, V. B. Prakapenka, and **S.-H. Shim**<sup>PI</sup>. Thermal expansion of SiC at high pressure–temperature and implications for thermal convection in the deep interiors of carbide exoplanets. *Journal of Geophysical Research: Planets*, 122, 124–133, DOI: 10.1002/2016JE005158 [↗](#), 2017.  
*Press release: EOS Research Spotlights* [↗](#) 
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4. **S.-H. Shim** and T. S. Duffy. Constraints on the P–V–T equation of state of  $\text{MgSiO}_3$  perovskite. *American Mineralogist*, 85, 354–363, DOI: 10.2138/am-2000-2-314 [↗](#), 2000.
3. **S.-H. Shim**, A. Navrotsky, T. R. Gaffney, and J. E. MacDougall. Chabazite: energetics of hydration, enthalpy of formation, and effect of cations on stability. *American Mineralogist*, 84, 1870–1882, DOI: 10.2138/am-1999-11-1214 [↗](#), 1999.
2. **S.-H. Shim**, S. J. Kim, and J. H. Ahn. Quantitative analysis of alkali feldspar minerals using Rietveld refinement of X-ray diffraction data. *American Mineralogist*, 81, 1133–1140, DOI: 10.2138/am-1996-9-1011 [↗](#), 1996.
1. **S.-H. Shim**, J. H. Ahn, and S. J. Kim. Quantitative analysis of feldspar mixture samples using the Rietveld refinement method. *Journal of the Mineralogical Society of Korea*, 7, 62–79, 1994.

## Peer-Reviewed Book Chapters

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1. **S.-H. Shim**. Stability of  $\text{MgSiO}_3$  perovskite in the lower mantle. In *Earth’s Deep Mantle: Structure, Composition, and Evolution*, edited by R. D. van der Hilst, J. Bass, J. Matas, and J. Trampert, volume 160 of *Geophysical Monograph Series*, 261–282. American Geophysical Union, 2005.

## Non-Peer-Reviewed Articles

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2. **S.-H. Shim** and T. Lay. Deep Earth: Post-perovskite at ten. *Nature Geoscience*, 7, 621–623, 2014.
1. T. Lay, D. Heinz, M. Ishii, **S.-H. Shim**, J. Tsuchiya, T. Tsuchiya, R. Wentzcovitch, and D. Yuen. Multidisciplinary impact of the deep mantle phase transition in perovskite structure. *Eos Transactions*, 86, 1–4, 2005.

## Apps, Jupyter Notebooks, and Codes

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Only the software and the codes with DOI are listed below. More software and codes can be found in Shim’s GitHub repository [↗](#). All the software and codes developed by Shim are open source.

10. **S.-H. Shim**. vibeels - A PyQt6 application for extracting and calibrating vibrational EELS spectra. Zenodo. DOI: 10.5281/zenodo.20018423 [↗](#), Github Repo [↗](#), 2026.
9. **S.-H. Shim**. vibfit - A Qt desktop application for constrained vibrational EELS peak fitting. Zenodo. DOI: 10.5281/zenodo.20018218 [↗](#), Github Repo [↗](#), 2026.
8. **X. Wei**<sup>GS</sup> and **S.-H. Shim**. IXE – X-ray emission spectroscopy analysis. Zenodo. DOI: 10.5281/zenodo.18346803 [↗](#), Github Repo [↗](#), 2026.
7. **S.-H. Shim**. Caker - A python application for batch processing diffraction images. Zenodo. DOI: 10.5281/zenodo.15015748 [↗](#), Github Repo [↗](#), 2025.
6. **S.-H. Shim**. Calculator for H in  $\text{FeH}_x$ . Zenodo. DOI: 10.5281/zenodo.7662676 [↗](#), Github Repo [↗](#), 2023.

5. **S.-H. Shim.** Inverse Birch-Murnaghan equation with spline interpolation. Zenodo. DOI: 10.5281/zenodo.6349563 [↗](#), Github Repo [↗](#), 2022.
4. **S.-H. Shim.** Unit cell fitting in Jupyter. Zenodo. DOI: 10.5281/zenodo.6344949 [↗](#), Github Repo [↗](#), 2022.
3. **S.-H. Shim.** JCPDSTools - A python app for creating, converting, and revising high-pressure diffraction information file. Zenodo. DOI: 10.5281/zenodo.6349449 [↗](#), Github Repo [↗](#), 2022.
2. **S.-H. Shim.** PeakPo - A python software for X-ray diffraction analysis at high pressure and high temperature. Zenodo. DOI: 10.5281/zenodo.810199 [↗](#), Github Repo [↗](#), 2017.
1. **S.-H. Shim.** Pytheos - a python tool set for equations of state. Zenodo. DOI: 10.5281/zenodo.802392 [↗](#), Github Repo [↗](#), 2017.

# Presentations

**Tags:** Shim's recent developments are highlighted with the following symbols.



Solar-system planets and exoplanets



Technical developments



Educational developments

## Colloquia and Lectures

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**2025:** Interdisciplinary Course “Warm Dense Matter: Concepts and Fundamentals”, virtual, Lawrence-Livermore National Laboratory



**2025:** Korea Basic Science Institute, Daejeon, South Korea



**2025:** High Pressure Geoscience Youth Forum in China, virtual



**2024:** Department of Earth and Environmental Sciences, University of Rochester, Rochester, NY



**2023:** Astrobiology Program, University of Washington, Seattle, WA



**2023:** Department of Earth, Environmental and Planetary Sciences, Brown University, Providence, RI



**2023:** Department of Geosciences, Princeton University, Princeton, NJ



**2023:** Ice Giant Systems Seminar Series, Applied Physics Laboratory, Johns Hopkins University, Webinar  
*Video on YouTube*, [↗](#)



**2023:** Department of Earth and Planetary Sciences, Stanford University, Stanford, CA



**2023:** High Energy Density Science Center, Livermore National Laboratory, Livermore, CA



**2022:** Advanced Photon Source High Pressure Special Interest Group Meeting, virtual



**2020:** Seismology Seminar, Seismology Laboratory, University of California, Berkeley, CA

**2020:** School of Earth and Space Exploration, Arizona State University, Tempe, AZ



**2019:** Department of Earth System Science, Yonsei University, Seoul, Korea



**2019:** School of Earth and Environmental Science, Seoul National University, Seoul, Korea

**2018:** China University of Geosciences, Wuhan, China

**2018:** School of Earth and Environmental Science, Seoul National University, Seoul, Korea

**2018:** Department of Earth System Sciences, Yonsei University, Seoul, Korea

**2018:** Equation of state, Cooperative Institute for Dynamic Earth Research (CIDER)–Kavli Institute for Theoretical Physics (KITP) Program: Relating Geophysical and Geochemical Heterogeneity in the Deep Earth, Santa Barbara, CA

**2017:** Mars InSight Mission team, Jet Propulsion Laboratory, CA



**2017:** Seismological Laboratory, California Institute of Technology, Pasadena, CA

**2016:** Public lecture on planetary interiors, KAOS foundation, Seoul, Korea



**2016:** School of Earth and Environmental Sciences, Seoul National University, Seoul, Korea

**2016:** Korea Basic Science Institute (KBSI), Chungju, Korea

**2016:** Lecture, Cooperative Institute for Dynamic Earth Research (CIDER), 2016 Summer Program, “Flow in the Deep Earth”, Santa Barbara, CA  
*Video Lecture*, [↗](#)

**2016:** Bavarian Research Institute of Experimental Geochemistry and Geophysics (BGI) University of

Bayreuth, Bayreuth, Germany

**2016:** Seismology Seminar, Seismology Laboratory, University of California, Berkeley, CA

**2014:** Department of Geology and Geophysics, University of Utah, Salt Lake City, UT

**2014:** Umbgrove Lectures, Department of Geosciences, Universiteit Utrecht, Utrecht, Nederland

**2014:** Institut de Physique du Globe de Paris (IPGP), Paris, France

**2013:** Center for the Origin, Dynamics and Evolution of Planets (CODEP), University of California, Santa Cruz, CA

**2012:** Institut de Physique du Globe de Paris (IPGP), Paris, France

**2011:** School of Environmental Science and Engineering, Pohang University of Science and Technology, Pohang, Korea

**2011:** School of Earth and Environmental Science, Seoul National University, Seoul, Korea

**2011:** Korea Astronomy and Space Science Institute, Daejeon, Korea

**2011:** School of Earth and Space Exploration, Arizona State University, Tempe, AZ

**2011:** Department of Geology, University of Maryland, College Park, MD

**2010:** Scripps Institution of Oceanography, University of California, San Diego, CA

**2010:** Department of Earth and Environmental Sciences, Ludwig-Maximilians University, Munich, Germany

**2010:** Department of Earth Sciences, Swiss Federal Institute of Technology (ETH), Zurich, Switzerland

**2010:** School of Earth and Environmental Science, Seoul National University, Seoul, Korea

**2010:** Department of Earth System Sciences, Yonsei University, Seoul, Korea

**2009:** Geophysics Colloquium, Princeton University, Princeton, NJ

**2008:** Department of Physics, University of Toronto, Toronto, Canada

**2007:** Department of Geological Sciences, Brown University, Providence, RI

**2007:** School of Earth and Environmental Science, Seoul National University, Seoul, Korea

**2007:** Department of Earth System Sciences, Yonsei University, Seoul, Korea

**2007:** Department of Earth and Environment Sciences, Chungbuk National University, Chungjoo, Korea

**2006:** Department of Earth and Atmospheric Sciences, Cornell University, Ithaca, NY

**2006:** Department of Earth and Environmental Sciences, Rensselaer Polytechnic Institute, Troy, NY

**2004:** Cooperative Institute for Deep Earth Research (CIDER) Summer Workshop, Kavli Institute for Theoretical Physics, University of California, Santa Barbara, CA

**2004:** Department of Earth and Planetary Sciences, Harvard University, Cambridge, MA

**2002:** School of Earth and Environmental Science, Seoul National University, Seoul, Korea

**2002:** Seismological Laboratory, University of California, Berkeley, CA

**2002:** Seismological Laboratory, California Institute of Technology, Pasadena, CA

**2001:** Department of Geology and Geophysics, Yale University, New Haven, CT

**2001:** Department of Geology, University of Illinois, Champaign, IL

**2001:** Department of Earth, Atmospheric, and Planetary Sciences, Massachusetts Institute of Technology, Cambridge, MA

## Invited Talks at Workshops

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- 2026:** Geological Society of Korea Planetary Science Short Course Season 5, “How to Study a Planet” 
- 2022:** Stanford Synchrotron Radiation Lightsource (SSRL) / Linac Coherent Light Source (LCLS) Users’ Meeting and Workshops, Menlo Park, CA
- 2021:** 10th Asian Conference on High Pressure Research, Virtual 
- 2021:** Advances in synchrotron-based research towards understanding the structure, evolution, and dynamics of Earth and planetary interiors, co-organized by Geoscience Synchrotron Users Group, GeoSoilEnviroCARS, and COMPRES, Virtual   
*Video on YouTube, [C3](#)*
- 2019:** Inelastic X-ray and Nuclear Resonant Scattering and Applications in Earth Sciences, Consortium for Materials Properties Research in Earth Sciences (COMPRES) Annual Meeting, Montana
- 2018:** Stanford Synchrotron Radiation Lightsource (SSRL) / Linac Coherent Light Source (LCLS) Users’ Meeting and Workshops, Menlo Park, CA
- 2018:** Synchrotron Infrared Spectroscopy on Materials in Extreme, National Synchrotron Light Source (NSLS)-II and Center for Functional Nanomaterials (CFN) Users’ Meeting, Brookhaven, NY
- 2018:** Advanced Photon Source / Center for Nanoscale Materials Users Meeting, Argonne, IL
- 2017:** 5th High power laser workshop, SLAC National Accelerator Laboratory, Menlo Park, CA
- 2016:** Deutsches Elektronen-Synchrotron – Arizona State University workshop, Hamburg, Germany
- 2016:** Institute for Study of the Earth’s Interior (ISEI) Misasa VI Symposium, Okayama University, Japan
- 2015:** Carbon at Extreme Conditions, Centre Européen de Calcul Atomique et Moléculaire (CECAM), CECAM-ETHZ, Lugano, Switzerland
- 2014:** PURE-4 Meeting (IPGP–UCL), Paris, France
- 2014:** PPv@10: A Meeting for the 10th Anniversary of the Discovery of Post-Perovskite, University of Bristol, Bristol, UK
- 2014:** Deep Earth Processes: Windows on the Working of a Planet, London, UK
- 2013:** Interior of the Earth, Gordon Research Conference, South Hadley, MA
- 2013:** Stellar Stoichiometry Workshop, Tempe, AZ
- 2012:** Structure and Dynamics of the Earth’s Deep Mantle, Collège de France, Paris, France
- 2010:** Computational Mineral Physics: Applications to Geophysics, Centre Européen de Calcul Atomique et Moléculaire (CECAM), CECAM-ETHZ, Zurich, Switzerland
- 2009:** Interior of the Earth, Gordon Research Conference, South Hadley, MA
- 2009:** Community Workshop, Cooperative Institute for Deep Earth Research, Marshall, CA
- 2008:** Asian Conference in High Pressure Research, Seoul, Korea
- 2008:** High-resolution X-ray Scattering on Earth Materials using Synchrotron Radiation, Advanced Photon Source (APS), Argonne National Laboratory, Argonne, IL
- 2008:** High-Pressure Workshop at the Joint National Synchrotron Light Source (NSLS) and Center for Functional Nanomaterials (CFN) Users Meeting, Brookhaven National Laboratory, Upton, NY
- 2006:** Workshop on Synergy of 21st Century High-Pressure Science and Technology Advanced Photon Source, Argonne National Laboratory, Argonne, IL
- 2005:** Virtual Laboratory for Earth and Planetary Materials (VLab) Workshop, University of Minnesota,

Minneapolis, MN

**2004:** Consortium for Materials Properties Research in Earth Sciences (COMPRES) Sponsored Workshop, Structure Determination by Single Crystal X-ray Diffraction at Megabar Pressures, Advanced Photon Source, Argonne National Laboratory, Argonne, IL

**2004:** GeoSoilEnviro Consortium for Advanced Radiation Source (GSECARS) / Consortium for Materials Properties Research in Earth Sciences (COMPRES) High-Pressure Workshop, Future Directions for the Laser-Heated Diamond Anvil Cell at the Advanced Photon Source, Argonne National Laboratory, Argonne, IL

## Invited Conference Presentations

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24. **S.-H. Shim**<sup>PI</sup>. From a Single Planet to Thousands of Worlds – Geoscience in the Age of Exoplanets. Geological Society of Korea, Fall Meeting, 2025, Jeju, South Korea. 
23. **S.-H. Shim**<sup>PI</sup>. Crucial Role of Materials Chemistry in Deciphering Exoplanet Composition, Structure, and Dynamics. Exoplanets in Our Backyard 2024, Louisville, KY. 
22. **S.-H. Shim**<sup>PI</sup>. High-Pressure Chemistry of Hydrogen and Water: Unveiling Implications for Mars, Earth, and Super-Earths. Gordon Research Conference – Research at High Pressure 2024, Holderness, NH. 
21. **S.-H. Shim**<sup>PI</sup>. High-Pressure Chemistry Linking Dry and Wet Planets. Abstract EE03.00001, 23rd Biennial Conference of the APS Topical Group on Shock Compression of Condensed Matter (SCCM23) Meeting 2023, Chicago, IL. 
20. **S.-H. Shim**<sup>PI</sup>. Hydrogen may play a key role for the fine-scale structures at the Earth's core-mantle boundary. International Union of Crystallography (IUCr) Meeting 2022, Chicago, IL.
19. **S.-H. Shim**<sup>PI</sup>, **C. Nisr**<sup>PD</sup>, **H. Piet**<sup>PD</sup>, **H. Chen**<sup>GS</sup>, **B. Ko**<sup>GS</sup>, **T. Kim**<sup>VG</sup>, **H. Horn**<sup>GS</sup>, S. Charison, V. Prakapenka, M. Kunz, Z. Liu, Y. Lee, A. Chizmeshya, J. O'Rourke, B. Chen, and S. Speziale. Hydrogen in the deep mantle Abstract 10059, Goldschmidt Meeting 2022, Honolulu, HI. 
18. **S.-H. Shim**<sup>PI</sup>, **H. Piet**<sup>PD</sup>, **S. Fu**<sup>PD</sup>, **B. Ko**<sup>GS</sup>, **T. Kim**<sup>VG</sup>, Y. Lee, V. B. Prakapenka, and S. Chariton. Impacts of hydrogen on the chemistry and structure of rocky planets' cores. Abstract DI23A-01, Fall American Geophysical Union (AGU) Meeting 2021, New Orleans, LA. 
17. **S.-H. Shim**<sup>PI</sup>, **H. Piet**<sup>PD</sup>, **H. Horn**<sup>GS</sup> and **B. Kulka**<sup>GS</sup>. Hydrogen-mineral reactions at high temperatures and high pressures. Abstract MR025-07, Fall American Geophysical Union (AGU) Meeting 2020, San Francisco, CA. 
16. **S.-H. Shim**<sup>PI</sup> and **J. Dolinski**<sup>GS</sup>. Exploring Earth and planets with large databases. Abstract ED044-0002, Fall American Geophysical Union (AGU) Meeting 2020, San Francisco, CA. 
15. **S.-H. Shim**<sup>PI</sup>, **B. Grocholski**<sup>PD</sup>, **Y. Ye**<sup>PD</sup>, E. E. Alp, Y. Meng, V. B. Prakapenka, D. Morgan, and S. Xu. Origin of the lower-mantle heterogeneities - importance of the crystal chemistry of bridgmanite. Abstract MR44A-02, Fall American Geophysical Union (AGU) Meeting 2017, New Orleans, LA.
14. **S.-H. Shim**<sup>PI</sup>, **C. Nisr**<sup>PD</sup>, M. Pagano, **H. Chen**<sup>GS</sup>, **B. Ko**<sup>GS</sup>, **S. Noble**<sup>UG</sup>, K. Leinenweber, P. Young, and S. Desch. Un-Earth-like interiors of the Earth-like planets. Abstract DI52B-01, Fall American Geophysical Union (AGU) Meeting 2015, San Francisco, CA. 

13. **S.-H. Shim**<sup>PI</sup>, **Y. Ye**<sup>PD</sup>, **B. Grocholski**<sup>PD</sup>, S. Xu, D. Morgan, J. Zhao, and E. Alp. The post-perovskite transition and mineralogical changes in the chemically heterogeneous lower mantle. Abstract MR23D-01, Fall American Geophysical Union (AGU) Meeting 2014, San Francisco, CA.
12. **S.-H. Shim**, R. D. van der Hilst, **B. Grocholski**<sup>PD</sup>, **K. Catalli**<sup>GS</sup>, Q. Cao, and X. Shang. Nature of mantle heterogeneities. Goldschmidt Conference 2011, The Geochemical Society, Prague, Czech Republic.
11. **S.-H. Shim**<sup>PI</sup>, **B. Grocholski**<sup>PD</sup>, **K. Catalli**<sup>GS</sup>, W. Sturhahn, and V. B. Prakapenka. Does the spin transition in mantle silicate perovskite change the seismic properties of the lower mantle? Abstract DI11B-03, Fall American Geophysical Union (AGU) Meeting 2010, San Francisco, CA.
10. **S.-H. Shim**<sup>PI</sup>, **B. Grocholski**<sup>PD</sup>, **K. Catalli**<sup>GS</sup>, W. Sturhahn, and V. B. Prakapenka. Iron in mantle silicate perovskite. Acta Mineralogica-Petrographica, Abstract Series, 6, 20th General Meeting of the International Mineralogical Association (IMA) 2010, Hungary, Budapest.
9. **S.-H. Shim**<sup>PI</sup>, **K. Catalli**<sup>GS</sup>, and **B. Grocholski**<sup>PD</sup>. Valence and spin states of iron in mantle silicate perovskite and their implications for the chemistry and mineralogy of the lower mantle. Abstract V23B-03, Eos Transactions American Geophysical Union, 91, Western Pacific Geophysics Meeting Supplement, 2010, Taipei, Taiwan.
8. **S.-H. Shim**<sup>PI</sup>, **K. Catalli**<sup>GS</sup>, **B. Grocholski**<sup>PD</sup>, and V. B. Prakapenka. Effect of compositional variation on the post-perovskite transition in the lowermost mantle. Japan Geoscience Union Meeting 2010, Chiba City, Japan.
7. **S.-H. Shim**<sup>PI</sup>, **K. Catalli**<sup>GS</sup>, and **B. Grocholski**<sup>PD</sup>. Mineralogical processes in the lower mantle and their implications for seismic heterogeneities. Abstract U23D-0066, Eos Transactions American Geophysical Union, 90, Fall Meeting Supplement, 2009, San Francisco, CA.
6. **S.-H. Shim**<sup>PI</sup>, **B. Grocholski**<sup>PD</sup>, and **K. Catalli**<sup>GS</sup>. Size matters - lessons from the interiors of Earth and Mars. Abstract P14B-02, Eos Transactions American Geophysical Union, 89, Fall Meeting Supplement, 2008, San Francisco, CA.
5. **S.-H. Shim**<sup>PI</sup>, **J. Hustoft**<sup>PD</sup>, **K. Catalli**<sup>GS</sup>, A. Kubo, V. B. Prakapenka, and M. Kunz. Bulk sound speed and Grüneisen parameter of postperovskite and their implications for the D'' heterogeneities. The 33rd International Geology Congress 2008, Oslo, Norway.
4. **S.-H. Shim**, T. S. Duffy, R. Jeanloz, and G. Shen. Stability and crystal structure of MgSiO<sub>3</sub> perovskite in the Earth's deep mantle. Abstract U34A-01, Eos Transactions American Geophysical Union, 85, Fall Meeting Supplement, 2004, San Francisco, CA.
3. **S.-H. Shim** and R. Jeanloz. Equation of state of MgSiO<sub>3</sub> perovskite and the chemical composition of the lower mantle. Abstract S338, Eos Transactions American Geophysical Union, 83, Spring Meeting Supplement, 2002, Washington, DC.
2. **S.-H. Shim**, T. S. Duffy, and G. Shen. The post-spinel phase boundary in Mg<sub>2</sub>SiO<sub>4</sub> and the 660-km seismic discontinuity. Abstract F889, Eos Transactions American Geophysical Union, 82, Fall Meeting Supplement, 2001, San Francisco, CA.
1. **S.-H. Shim**, T. S. Duffy, and G. Shen. In situ determination of phase boundary between spinel and perovskite+periclase in Mg<sub>2</sub>SiO<sub>4</sub> by laser heated diamond anvil cell. Abstract F1207, Eos Transactions American Geophysical Union, 81, Fall Meeting Supplement, 2000, San Francisco, CA.

## Contributed Conference Presentations

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206. **A. Pease**<sup>PD</sup>, **S. Chen**<sup>PD</sup>, **T. Kim**<sup>PD</sup>, and **S.-H. Shim**<sup>PI</sup>. Applying TWISTER to Study Planetary Cores under Static and Dynamic Compression. SEES-FORCE 2026 Joint Meeting, Chicago, IL, August 10–12, 2026.  
205. **A. Pease**<sup>PD</sup>, K. Amezcua, and **S.-H. Shim**<sup>PI</sup>. EoSAlign: An Open-Source Software for Calculating, Comparing, and Aligning Pressure under Extreme Conditions. Poster 26, SEES-FORCE 2026 Joint Meeting, Chicago, IL, August 10–12, 2026. 
204. **A. Tukić**<sup>GS</sup>, V. Prakapenka, S. Chariton, and **S.-H. Shim**<sup>PI</sup>. Hydrogen-Silicate Reactions at Low H<sub>2</sub> Availability: Implications for Water Production in Atmosphere-Depleted Sub-Neptunes. Poster 40, SEES-FORCE 2026 Joint Meeting, Chicago, IL, August 10–12, 2026. 
203. **S.-H. Shim**<sup>PI</sup>, **H. Horn**<sup>GS</sup>, **T. Kim**<sup>PD</sup>, **X. Wei**<sup>PD</sup>, A. Vazan, V. Prakapenka, and S. Chariton. Chemical Alteration of Sub-Neptune Atmospheres by Reactions with Molten Silicate Interiors. 11th AAS Topical Conference Series: Exoplanet Atmospheres 2026, Denver, CO, March 16–20, 2026. 
202. **T. Kim**<sup>PD</sup>, A. Vazan, S. Chariton, V. Prakapenka, Y.-J. Ryu, S. Yang, and **S.-H. Shim**<sup>PI</sup>. Chemical coupling between atmosphere and interior of sub-Neptunes: insights from high-pressure experiments. Abstract 443.23, 247th Meeting of the American Astronomical Society 2026, Phoenix, AZ.
201. Y. Zhang, K. Bali, M. Andronaco, H. Yang, A. Chen, L. Libon, E. Boulard, A. Benuzzi-Mounaix, E. Galtier, R. Caracas, A. E. Gleason, G. Morard, C. Dorn, A. Ravasio, S. Pandolfi, **X. Wei**<sup>PD</sup>, D. Zheng, C. Prescher, H. J. Lee, N. Czapla, R. Alonso-Mori, **S.-H. Shim**, and W. L. Mao. Laser-driven shock compression on FeOOH: Implications for Fe-O-H interaction in a magma ocean. Abstract MR12A-01, Fall American Geophysical Union (AGU) Meeting 2025, New Orleans, LA.
200. **A. Pease**<sup>PD</sup>, H. Krauss, and **S.-H. Shim**<sup>PI</sup>. An Open-Source Software for Calculating and Comparing Pressure Under Extreme Conditions. Abstract MR12A-09, Fall American Geophysical Union (AGU) Meeting 2025, New Orleans, LA.
199. **S. Chen**<sup>PD</sup>, **T. Kim**<sup>PD</sup>, M. Bose, S. Chariton, V. Prakapenka, D. Smith, R. Hrubiak, and **S.-H. Shim**<sup>PI</sup>. Hydrogen at the core-mantle boundary of Mars. Abstract DI51A-05, Fall American Geophysical Union (AGU) Meeting 2025, New Orleans, LA.
198. **X. Wei**<sup>GS</sup>, **T. Kim**<sup>PD</sup>, **S. Chen**<sup>PD</sup>, S. Chariton, V. B. Prakapenka, and **S.-H. Shim**<sup>PI</sup>. Equation of State of Mg<sub>2</sub>FeH<sub>6</sub> Perovskite: Implications for Hydrogen Ingassing in Sub-Neptune Interiors. Abstract DI51A-04, Fall American Geophysical Union (AGU) Meeting 2025, New Orleans, LA.
197. **A. Campbell**<sup>US</sup>, **S. Fu**<sup>PD</sup>, L. Shteynman, **S. Chen**<sup>PD</sup>, **T. Kim**<sup>PD</sup>, and **S.-H. Shim**<sup>PI</sup>. Hydrogen Production and Storage in CaTiO<sub>3</sub> Perovskite at Mantle-Related High Pressures. Abstract DI41A-0003, Fall American Geophysical Union (AGU) Meeting 2025, New Orleans, LA.
196. **S. Chen**<sup>PD</sup>, **A. M. Pease**<sup>PD</sup>, I. Szumila, S. J. Tracy, A. Tıpeev, I. Oleynik, and **S.-H. Shim**<sup>PI</sup>. Dynamic Compression of Iron-Sulfur Alloys at the Earth's Core Conditions. Abstract DI33B-0025, Fall American Geophysical Union (AGU) Meeting 2025, New Orleans, LA.
195. J. Zhang, W.-Y. Zhou, M. Hao, W. Su, **T. Kim**<sup>PD</sup>, **S. Chen**<sup>PD</sup>, **S.-H. Shim**, P. Nguyen, D. Zhang, and K. Armstrong. Elasticity of Fe,Al,Mg,Ti-bearing Davemaoite Explains Lower Mantle Seismic Heterogeneities. Abstract DI22A-03, Fall American Geophysical Union (AGU) Meeting 2025, New Orleans, LA.

194. **T. Kim**<sup>PD</sup>, Y. Lee, S. Chariton, V. Prakapenka, E. Greenberg, and **S.-H. Shim**<sup>PI</sup>. Effects of Water on the Post-perovskite Transition. Abstract DI23A-07, Fall American Geophysical Union (AGU) Meeting 2025, New Orleans, LA.
193. **T. Kim**<sup>PD</sup>, A. Vazan, S. Chariton, V. Prakapenka, Y. J. Ryu, S. Yang, and **S.-H. Shim**<sup>PI</sup>. Impact of increasing H<sub>2</sub>O concentration during conversion of a hydrogen-rich sub-Neptune to a water-rich sub-Neptune. Fall American Geophysical Union (AGU) Meeting 2025, New Orleans, LA.
192. **S.-H. Shim**, H. Horn, T. Kim, X. Wei, and A. Vazan. Hydrogen-magma reactions provide a new mechanism for generating substantial water inventories in sub-Neptune exoplanets. Gordon conference 2025, South Hadley, MA.
191. **S. Chen**<sup>PD</sup>, **T. Kim**<sup>PD</sup>, **X. Wei**<sup>GS</sup>, and **S.-H. Shim**<sup>PI</sup>. Partitioning of Potassium Between FeO and Fe Metal Impacted by Hydrogen and Implications for Earth's Core. Abstract DI13A-3153, Fall American Geophysical Union (AGU) Meeting 2024, Washington, DC.
190. **X. Wei**<sup>GS</sup>, **S.-H. Shim**<sup>PI</sup>, H. J. Lee, E. Galtier, R. Alonso-Mori, H. Yang, S. Park, W. L. Mao, A. E. Gleason, S. Glenzer, A. Ravasio, S. Pandolfi, G. Morard, and J.-A. Hernandez. Dynamic Compression Study for Fe Spin States in Silicate Melts under Pressure-Temperature Conditions of Super-Earth's Deep Interiors. Abstract DI13B-07, Fall American Geophysical Union (AGU) Meeting 2024, Washington, DC.
189. **T. Kim**<sup>PD</sup>, **S.-H. Shim**<sup>PI</sup>, S. Chariton, V. Prakapenka, M. Kunz and K. Armstrong. Iron metal formation from silicate melts at 55–75 GPa and 4000–6000 K. Abstract DI31C-2947, Fall American Geophysical Union (AGU) Meeting 2024, Washington, DC.
188. **T. Kim**<sup>PD</sup>, **S.-H. Shim**<sup>PI</sup>, M. Li, E. Garnerio, E. Greenberg, S. Chariton, V. Prakapenka, S. Yang, and Y. Lee. Understanding the ultralow velocity zones at the Earth's core-mantle boundary through synchrotron experiments. Abstract MR21A-07, Fall American Geophysical Union (AGU) Meeting 2024, Washington, DC.
187. T. Sharp, **S.-H. Shim**, K. Leinenweber, D. C. Smith, A. Navrotsky, K. Brugman, S. Chen, K. Tsuno, and L. Leinbach. FORCE (Facility for Open Research in a Compressed Environment) Progress, instrumentation and Initial Experiments. Abstract MR31B-2986, Fall American Geophysical Union (AGU) Meeting 2024, Washington, DC.
186. Y. Zhang, A. Ravasio, H. Yang, S. Pandolfi, A. Chen, L. Libon, E. Boulard, A. E. Gleason, **S.-H. Shim**, G. Morard, and W. L. Mao. Hugoniot Equation of State of FeOOH up to 818 GPa: Insights into Magma Oceans in Exoplanets. Abstract DI11B-3133, Fall American Geophysical Union (AGU) Meeting 2024, Washington, DC.
185. A. J. Chen, H. Yang, Y. Zhang, A. E. Gleason, R. Alonso-Mori, G. Morard, A. Ravasio, **S.-H. Shim**, **X. Wei**<sup>GS</sup>, E. Galtier, Y. Han, H. J. Lee, B. Nagler, D. Sokaras, E. Boulard, S. Pandolfi, Q. Hu, J.-A. Hernandez, B. Cui, X. Chen, and W. L. Mao. Probing Atomic and Electronic Structures of Laser-Shock Compressed Hematite ( $\alpha$ -Fe<sub>2</sub>O<sub>3</sub>) to 300 GPa. Abstract DI11B-3148, Fall American Geophysical Union (AGU) Meeting 2024, Washington, DC.
184. **T. Kim**<sup>PD</sup> and **S.-H. Shim**<sup>PI</sup>. High-pressure experiments support compositional gradients in sub-Neptune interiors. Exoplanets in Our Backyard 2024, Louisville, KY.
183. K. Tsuno, D. S. Grewal, V. Xu, L. Leinbach, K. Leinenweber, A. Wittmann, and **S.-H. Shim**. Wetting property of Fe-Ni-N alloy melt in ringwoodite: Implications for nitrogen depletion in the bulk silicate Earth. Goldschmidt Conference 2024, Chicago, IL.

182. **X. Wei**<sup>GS</sup>, **S.-H. Shim**, H. J. Lee, E. Galtier, R. Alonso-Mori, H., Yang, W. L. Mao, A. E. Gleason, S. Glenzer, J.-A. Hernandez, A. Ravasio, G. Morard, S. Pandolfi. Spin State Behavior of Iron in Molten Silicates under Dynamic Compression: Implications for Super-Earths. Ultrafast X-Ray Summer School 2024, SLAC National Lab, CA. 
181. K. Leinenweber, **S.-H. Shim**, and A. Navrotsky. A Nancy Ross-style description of high-pressure hydrous stishovite ((Si,H<sub>4</sub>)O<sub>2</sub>). Geological Society of America Meeting 2024, Anaheim, CA.
180. **S.-H. Shim**<sup>PI</sup>, T. Kim, X. Wei, S. Chariton, V. Prakapenka, Y.-J. Ryu, and S. Yang. Formation of hydride perovskite and water on hot hydrogen-rich exoplanets. Geological Society of America Meeting 2024, Anaheim, CA. 
179. **S. Chen**<sup>PD</sup>, S. Yang, T. Kim, X. Wei, and **S.-H. Shim**<sup>PI</sup>. Sub-nanometer detection of hydrogen in minerals using vibrational electron energy loss spectroscopy. Geological Society of America Meeting 2024, Anaheim, CA.
178. **X. Wei**<sup>GS</sup>, **S.-H. Shim**, H. J. Lee, E. Galtier, R. Alonso-Mori, H. Yang, W. L. Mao, A. E. Gleason, S. Glenzer, J.-A. Hernandez, A. Ravasio, G. Morard, S. Pandolfi. Spin State of Fe in Dynamically Compressed Molten Silicates: Insights into Super-Earths' Deep Interiors. Gordon conference 2024, Holderness, NH.
177. **T. Kim**<sup>PD</sup>, **X. Wei**<sup>GS</sup>, S. Chariton, V. Prakapenka, Y. J. Ryu, S. Yang, and **S.-H. Shim**<sup>PI</sup>. Formation of Mg-hydrides and water from reaction between MgO and hydrogen and implication for sub-Neptune exoplanets. Abstract DI51A-07, Fall American Geophysical Union (AGU) Meeting 2023, San Francisco, CA. 
176. K. J. Agboola, S. Hansen, E. Garnero, M. Li, **S.-H. Shim**, and S. Rost. Ubiquitous ultra-low velocity zone structure beneath the southern hemisphere: evidence from double-array stacking of *PcP* waveforms. Abstract DI51B-0016, Fall American Geophysical Union (AGU) Meeting 2023, San Francisco, CA.
175. L. Leinbach, K. Brugman, K. Leinenweber, A. Navrotsky, T. G. Sharp, **S.-H. Shim**, D. J. Smith, and K. Tsuno. FORCE (Facility for Open Research in a Compressed Environment) progress and future advances. Abstract MR21B-0049, Fall American Geophysical Union (AGU) Meeting 2023, San Francisco, CA.
174. **H. Horn**<sup>GS</sup>, S. Chariton, V. Prakapenka, and **S.-H. Shim**<sup>PI</sup>. Reduction of silicates by hydrogen at high pressure-temperature – Experimental evidence for the formation of hot wet sub-Neptunes. Abstract MR24A-07, Fall American Geophysical Union (AGU) Meeting 2023, San Francisco, CA. 
173. **S. Chen**<sup>PD</sup>, S. Yang, **T. Kim**<sup>PD</sup>, **X. Wei**<sup>GS</sup>, and **S.-H. Shim**<sup>PI</sup>. Nondestructive sub-nanometer probe for hydrogen in high-pressure minerals: vibrational electron energy loss spectroscopy. Abstract MR32A-07A, Fall American Geophysical Union (AGU) Meeting 2023, San Francisco, CA.
172. **X. Wei**<sup>GS</sup>, S. Chariton, V. Prakapenka, M. Kunz, K. Armstrong, S. Yang, and **S.-H. Shim**<sup>PI</sup>. Effects of H and Si on the solubility of Mg in the metallic iron core, Abstract MR43C-0098, Fall American Geophysical Union (AGU) Meeting 2023, San Francisco, CA. 
171. **T. Kim**<sup>PD</sup>, S. Chariton, V. Prakapenka, Y. J. Ryu, S. Yang, and **S.-H. Shim**<sup>PI</sup>. Effect of H<sub>2</sub>-to-H<sub>2</sub>O ratio on the stability of silicates in sub-Neptune exoplanets. Abstract P23F-3106, Fall American Geophysical Union (AGU) Meeting 2023, San Francisco, CA. 
170. K. Tsuno, V. Xu, D. S. Grewal, L. Leinbach, K. Leinenweber, A. Wittmann, and **S.-H. Shim**. Wetting property of Fe-Ni-N alloy liquid in ringwoodite – implications for nitrogen depletion in the bulk silicate

Earth. Abstract V11C-0076, Fall American Geophysical Union (AGU) Meeting 2023, San Francisco, CA.

169. **H. W. Horn**<sup>GS</sup>, V. Prakapenka, S. Chariton, and **S.-H. Shim**<sup>PI</sup>. Reduction of Silicates by Hydrogen at High Pressure-Temperature - Experimental Evidence for the Formation of Hot Wet Sub-Neptunes. Abstract EE03.00002, 23rd Biennial Conference of the APS Topical Group on Shock Compression of Condensed Matter (SCCM23) Meeting 2023, Chicago, IL. 
168. **S.-H. Shim**<sup>PI</sup>. Endogenic Water from Hydrogen-Magma Reaction in Sub-Neptunes — Implications for Atmospheric Composition, Internal Structure, and Demographics. Ninth Annual Giant Magellan Telescope Community Science Meeting 2023, Washington, DC. 
167. **X. Wei**<sup>GS</sup>, V. Prakapenka, S. Chariton, S. Yang, and **S.-H. Shim**<sup>PI</sup>. How Hydrogen Could Facilitate Storage of Magnesium in Iron and Power an Early Dynamo. Gordon conference 2023, Mount Holyoke College, MA. 
166. **X. Wei**<sup>GS</sup>, S. Chariton, V. Prakapenka, and **S.-H. Shim**<sup>PI</sup>. Phase Relations of Fe-S-H for the Martian Core. International Union of Crystallography (IUCr) Meeting 2022, Chicago, IL. 
165. **S. Ravikumar**<sup>US</sup> and **S.-H. Shim**<sup>PI</sup>. A Potential Mechanism for Nitrogen Storage in the Earth's Mantle Transition Zone. International Union of Crystallography (IUCr) Meeting 2022, Chicago, IL.
164. **T. Kim**<sup>PD</sup>, E. Greenberg, V. B. Prakapenka, Y. Lee, and **S.-H. Shim**<sup>PI</sup>. Melting behaviors of hydrous pyrolite in the lower mantle. Abstract DI22A-09, Fall American Geophysical Union (AGU) Meeting 2022, Chicago, IL.
163. **B. Ko**<sup>GS</sup>, E. Greenberg, V. B. Prakapenka, E. Alp, W. Bi, Y. Meng, D. Zhang, and **S.-H. Shim**<sup>PI</sup>. Dissolution of calcium into bridgmanite and disappearance of davemaioite in warm regions of the lower mantle. Abstract DI26A-02, Fall American Geophysical Union (AGU) Meeting 2022, Chicago, IL.
162. **T. Kim**<sup>PD</sup>, J. O'Rourke, J. Lee, S. Chariton, V. Prakapenka, R. Husband, N. Giordano, H.-P. Liermann, Y. Lee, and **S.-H. Shim**<sup>PI</sup>. Possible link between deeply subducted water and the *E'* layer. Abstract DI36A-05, Fall American Geophysical Union (AGU) Meeting 2022, Chicago, IL.
161. **X. Wei**<sup>GS</sup>, S. Chariton, V. Prakapenka, and **S.-H. Shim**<sup>PI</sup>. Effects of Hydrogen on Fe-S Alloy System and their Implications for the Martian Core. Abstract MR35B-0059, Fall American Geophysical Union (AGU) Meeting 2022, Chicago, IL.
160. K. Tsuno, K. Leinenweber, A. Navrotsky, T. Sharp, **S.-H. Shim**, and D. Smith. New High-Pressure Experimental facility in FORCE (Facility for Open Research in a Compressed Environment) and scientific applications. Abstract MR42A-0066, Fall American Geophysical Union (AGU) Meeting 2022, Chicago, IL.
159. S. Hansen, E. Garnero, M. Li, **S.-H. Shim**, and S. Rost. A Global Veneer of Subducted Materials along the Earth's Core-Mantle Boundary. Abstract DI22B-0002, Fall American Geophysical Union (AGU) Meeting 2022, Chicago, IL.
158. K. Leinenweber, A. Navrotsky, D. Smith, T. Sharp, **S.-H. Shim**, K. Tsuno, K. Brugman. FORCE – A Newly Created “Facility for Open Research in a Compressed Environment.” Abstract MR45A-03, Fall American Geophysical Union (AGU) Meeting 2022, Chicago, IL.
157. **S.-H. Shim**<sup>PI</sup>, **H. Horn**<sup>GS</sup>, S. Speziale, S. Chariton, and V. Prakapenka. Chemistry Linking Hydrogen-Rich and Water-Rich Planets. Abstract 3031, Exoplanets in Our Backyard 2, 2022, Albuquerque, NM. 

156. **T. Kim**<sup>PD</sup>, **X. Wei**<sup>GS</sup>, S. Chariton, V. Prakapenka, **S.-H. Shim**<sup>PI</sup>. Impact of the H<sub>2</sub>/H<sub>2</sub>O Ratio on the Internal Structure of Uranus, Neptune, and Sub-Neptune Exoplanets. Abstract 3052, Exoplanets in Our Backyard 2, 2022, Albuquerque, NM. 
155. **S.-H. Shim**<sup>PI</sup>, **H. Horn**<sup>GS</sup>, S. Speziale, S. Chariton, and V. Prakapenka. Hydrogen chemistry for understanding sub-Neptunes. Abstract 404.05, Meeting of the AAS Division for Planetary Sciences 2022, London, Canada. 
154. **T. Kim**<sup>PD</sup>, Y. Lee, S. Chariton, V. Prakapenka, A. Pakhomova, H.-P. Liermann, Z. Liu, S. Speziale, and **S.-H. Shim**<sup>PI</sup>. Solubility of MgO in H<sub>2</sub>O-H<sub>2</sub> at high pressures and its implications for large water-rich planets. Abstract 107.03, Meeting of the AAS Division for Planetary Sciences 2022, London, Canada. 
153. **S. Fu**<sup>PD</sup>, S. Chariton, V. B. Prakapenka, A. Chizmeshya, and **S.-H. Shim**<sup>PI</sup>. Phase Relations in the Fe-Si-H Ternary up to 125 GPa and 3700 K. Abstract EGU22-3229, European Geophysical Union (EGU) Meeting 2022, Vienna, Austria and Online.
152. **S. Fu**<sup>PD</sup>, S. Chariton, V. B. Prakapenka, and **S.-H. Shim**<sup>PI</sup>. Melting Behavior of Fe-Si-H and its Impact on Possible Stratification at the Topmost Outer Core. Goldschmidt Meeting 2022, Honolulu, HI.
151. **H. Horn**<sup>GS</sup>, S. Chariton, V. B. Prakapenka, and **S.-H. Shim**<sup>PI</sup>. Chemical interactions between hydrogen-rich envelopes and silicate magma oceans. Abstract MR15A-0045, Fall American Geophysical Union (AGU) Meeting 2021, New Orleans, LA.
150. Q. Yuan, M. Li, E. Garnero, **S.-H. Shim**, and **B. Ko**<sup>GS</sup>. Investigating the dynamics of ultra-high velocity zones above Earth's core-mantle boundary. Abstract DI44B-08, Fall American Geophysical Union (AGU) Meeting 2021, New Orleans, LA.
149. **T. Kim**<sup>VG</sup>, S. Chariton, V. B. Prakapenka, A. Pakhomova, H.-P. Liermann, Z. Liu, S. Speziale, **S.-H. Shim**<sup>PI</sup>, and Y. Lee. Atomic-scale mixing between MgO and H<sub>2</sub>O in the deep interiors of water-rich planets. Abstract DI32A-06, Fall American Geophysical Union (AGU) Meeting 2021, New Orleans, LA.
148. **S. Fu**<sup>PD</sup>, S. Chariton, V. B. Prakapenka, A. Chizmeshya, and **S.-H. Shim**<sup>PI</sup>. Alloys in Fe-Si-H ternary at high pressures-temperatures and their implications for the planetary cores. Abstract DI25C-0047, Fall American Geophysical Union (AGU) Meeting 2021, New Orleans, LA.
147. S. E. Hansen, E. Garnero, S. Rost, M. Li, and **S.-H. Shim**. Ultra-low velocity zone structure at southern latitudes based on *PcP* waveforms and historical interstation pattern referencing. Abstract DI14A-09, Fall American Geophysical Union (AGU) Meeting 2021, New Orleans, LA.
146. **S.-H. Shim**<sup>PI</sup>, R. Alonso-Mori, A. Gleason, W. Mao, G. Morard, A. Ravasio, **B. Ko**<sup>GS</sup>, **J. Dolinski**<sup>GS</sup>, **J. Tappan**<sup>UG</sup>, D. Sokaras, B. Nagler, H. J. Lee, E. Galtier, R. Bolis, J.-A. Hernandez, M. Guarguaglini, A. Benuzzi-Mounaix, T. Vinci, G. Fiquet, and M. Baron. Spin state of iron in dynamically compressed olivine melt. Abstract MR12A-08, Fall American Geophysical Union (AGU) Meeting 2021, New Orleans, LA.
145. **T. Kim**<sup>VG</sup>, J. Lee, S. Chariton, V. B. Prakapenka, R. Husband, N. Giordano, H.-P. Liermann, and **S.-H. Shim**<sup>PI</sup>. Water may limit silicon amount in the Earth's core. Abstract 07-0934, 10th Asian Conference on High Pressure Research 2021.
144. **S.-H. Shim**<sup>PI</sup>. Solubility of magnesium in water at high pressures and its implications for Uranus. Abstract 314.03, Meeting of the AAS Division for Planetary Sciences 2021. 

143. **T. Kim**<sup>VGs</sup>, S. Chariton, V. Prakapenka, A. Pakhomova, H.-P. Liermann, Z. Liu, S. Speziale, **S.-H. Shim**<sup>PI</sup>, and Y. J. Lee Atomic-scale mixing between MgO and H<sub>2</sub>O in the deep interiors of water-rich planets. Abstract 495, Europlanet Science Congress 2021, Virtual. 
142. **S.-H. Shim**, **C. Nisr**<sup>PD</sup>, **T. Kim**<sup>VGs</sup>, Y. J. Lee, A. Chizmeshya, K. Leinenweber, S. Chariton, V. Prakapenka, S. Speziale, Z. Liu, and H.-P. Liermann. Mineral-Water Reaction at High Pressures - Implications for Uranus and Neptune. Abstract 1208, Lunar and Planetary Science Conference 2021, Houston, TX. 
141. **S. Fu**<sup>PD</sup> and **S.-H. Shim**<sup>PI</sup>. Hydrogen Storage in FeSi Alloy at High Pressure: Implications for the Composition and Structure of Rocky Planets' Cores. Abstract 1735, Lunar and Planetary Science Conference 2021, Houston, TX.
140. **S. Fu**<sup>PD</sup>, K. Leinenweber, **S.-H. Shim**. Potential storage of molecular hydrogen in CaTiO<sub>3</sub> perovskite in the deep interiors of rocky planets. AASTCS 8: Habitable Worlds, 2021, Virtual.
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137. **H. Horn**<sup>GS</sup>, V. B. Prakapenka, S. Chariton, S. Speziale, and **S.-H. Shim**<sup>PI</sup>. Reduction of iron by hydrogen in early planetary mantles. Abstract DI005-0007, Fall American Geophysical Union (AGU) Meeting 2020, San Francisco, CA. 
136. **B. Ko**<sup>GS</sup>, S. Chariton, V. B. Prakapenka, B. Chen, S. Yu, E. Garner, M. Li, and **S.-H. Shim**<sup>PI</sup>. Water-induced diamond formation at the Earth's core-mantle boundary. Abstract DI005-0023, Fall American Geophysical Union (AGU) Meeting 2020, San Francisco, CA.
135. E. Garner, S. Yu, **S.-H. Shim**, M. Li, **B. Ko**<sup>GS</sup>, M. S. Thorne, and C. Zhao. Ultra high velocity zones at the core-mantle boundary. Abstract DI008-05, Fall American Geophysical Union (AGU) Meeting 2020, San Francisco, CA.
134. **H. Horn**<sup>GS</sup> and **S.-H. Shim**<sup>PI</sup>. Effects of water on the mineralogy of carbon-rich exoplanets. Annual Meeting of Consortium for Materials Properties Research in Earth Sciences (COMPRES) 2020. 
133. **B. Kulka**<sup>GS</sup> and **S.-H. Shim**<sup>PI</sup>. Toward oxygen fugacity control in laser-heated diamond-anvil cell. Annual Meeting of Consortium for Materials Properties Research in Earth Sciences (COMPRES) 2020.
132. **B. Ko**<sup>GS</sup> and **S.-H. Shim**<sup>PI</sup>. Temperature-dependent solubility of uranium in silicate perovskites in the Earth's lower mantle. Annual Meeting of Consortium for Materials Properties Research in Earth Sciences (COMPRES) 2020.
131. **S.-H. Shim**<sup>PI</sup>. Superstoichiometric alloying of FeNi with H and hydrogen storage in the interiors of giant planets. Abstract 216.01, Meeting of the AAS Division for Planetary Sciences 2020. 
130. **J. G. O'Rourke**<sup>PD</sup> and **S.-H. Shim**<sup>PI</sup>. Hydrogenation of the Martian core by hydrated mantle minerals with implications for the early dynamo. Abstract 2475, Lunar and Planetary Science Conference 2020, Houston, TX. 

129. **H. Piet**<sup>PD</sup>, K. Leinenweber, E. Greenberg, S. Chariton, V. B. Prakapenka, P. R. Buseck, and **S.-H. Shim**<sup>PI</sup>.  A potential stratification of the core of Mars caused by hydrogen. Abstract 3025, Exoplanets in our Backyard Meeting 2020, Houston, TX.
128. **S.-H. Shim**<sup>PI</sup>, **C. Nisr**<sup>PD</sup>, **T. Kim**<sup>VG</sup>, Y. Lee, **H. Chen**<sup>GS</sup>, K. Leinenweber, A. V. G. Chizmeshya, S. Speziale, V. B. Prakapenka, C. Prescher, S. Tkachev, Y. Meng, and Z. Liu.  Experimental Observations on water-rock interaction at high pressures and their implications for the interiors of Uranus and Neptune. Abstract 3027, Exoplanets in our Backyard Meeting 2020, Houston, TX.
127. **T. Kim**<sup>VG</sup>, **S.-H. Shim**<sup>PI</sup>, V. B. Prakapenka, H.-P. Liermann, S. Speziale, Y. Lee.  High solubility of Mg in H<sub>2</sub>O at high pressures and its implications for the interiors of water-rich planets. Abstract 3031, Exoplanets in our Backyard Meeting 2020, Houston, TX.
126. **H. Horn**<sup>GS</sup>, S. Speziale, V. B. Prakapenka, and **S.-H. Shim**<sup>PI</sup>.  Interaction between iron/magnesium oxides and hydrogen. Abstract 3036, Exoplanets in our Backyard Meeting 2020, Houston, TX.
125. C. M. Lisse, S. J. Desch, C. T. Unterborn, S. R. Kane, P. R. Young, H. E. Hartnett, N. R. Hinkel, **S.-H. Shim**, and E. E. Mamajek.  Procedure for observing rocky exoplanets to maximize the likelihood of atmospheric oxygen biosignatures. Abstract 3064, Exoplanets in our Backyard Meeting 2020, Houston, TX.
124. **S.-H. Shim**<sup>PI</sup>, **H. Piet**<sup>PD</sup>, **J. O'Rourke**<sup>PD</sup>, K. Leinenweber, V. B. Prakapenka, E. Greenberg, and P. Buseck.  Possible immiscibility of iron hydride and iron sulfide in the Martian core. Abstract EPSC-DPS2019-188, European Planetary Science Congress - Division for Planetary Science Joint Meeting 2019, Geneva, Switzerland.
123. **S.-H. Shim**<sup>PI</sup>, **C. Nisr**<sup>PD</sup>, **H. Chen**<sup>GS</sup>, K. Leinenweber, A. V. G. Chizmeshya, V. B. Prakapenka, C. Prescher, S. Tkachev, Y. Meng, and Z. Liu.  Breaking the barrier between silica and water in the deep interiors of water-world exoplanets. Abstract EPSC-DPS2019-189, European Planetary Science Congress - Division for Planetary Science Joint Meeting 2019, Geneva, Switzerland.
122. **H. Piet**<sup>PD</sup>, K. Leinenweber, E. Greenberg, V. B. Prakapenka, P. R. Buseck, and **S.-H. Shim**<sup>PI</sup>.  Hydrogen and sulfur in metallic iron in the core of Mars. Annual Meeting of Consortium for Materials Properties Research in Earth Sciences (COMPRES) 2019, Big Sky, MT.
121. **H. Chen**<sup>GS</sup> and **S.-H. Shim**<sup>PI</sup>.  Mineralogy of the hydrous lower mantle. Annual Meeting of Consortium for Materials Properties Research in Earth Sciences (COMPRES) 2019, Big Sky, MT.
120. **H. Horn**<sup>GS</sup> and **S.-H. Shim**<sup>PI</sup>.  Oxidation of the interiors of carbide exo-planets. Abstract 341-354, Astrobiology Science Conference (AbSciCon) 2019, Seattle, WA.
119. **J. Dolinski**<sup>GS</sup> and **S.-H. Shim**<sup>PI</sup>.  Teaching resource in Jupyter notebooks for accessing and analyzing large research databases in Earth and planetary science. Abstract ED53F-0904, Fall American Geophysical Union (AGU) Meeting 2019, San Francisco, CA.
118. **J. Dolinski**<sup>GS</sup>, K. Leinenweber, V. B. Prakapenka, E. Greenberg, D. Zhang, A. Wittmann, W. Bi, E. Alp, and **S.-H. Shim**<sup>PI</sup>.  Effects of Mg/Si ratio on the mineralogy of the Martian mantle constrained by high-pressure experiments. Abstract DI42A-07, Fall American Geophysical Union (AGU) Meeting 2019, San Francisco, CA.
117. **H. Horn**<sup>GS</sup>, K. Leinenweber, V. B. Prakapenka, E. Greenberg, and **S.-H. Shim**<sup>PI</sup>.  Chemical reaction between silicon carbide and water and its implication for the stability of carbide exo-planets. Abstract DI41C-0017, Fall American Geophysical Union (AGU) Meeting 2019, San Francisco, CA.

116. **B. Ko**<sup>GS</sup>, **S.-H. Shim**<sup>PI</sup>, E. Greenberg, and V. B. Prakapenka. The solubility of uranium in bridgmanite in the Earth's lower mantle. Abstract DI41C-0017, Fall American Geophysical Union (AGU) Meeting 2019, San Francisco, CA.
115. **T. Kim**<sup>VG</sup>, **S.-H. Shim**, B. Ko, E. Greenberg, V. B. Prakapenka, and Y. Lee. Low melting temperatures of anhydrous and hydrous mantle materials at the core-mantle boundary. Abstract DI41C-0017, Fall American Geophysical Union (AGU) Meeting 2019, San Francisco, CA.
114. C. Millsaps, M. Li, C. Unterborn, and **S.-H. Shim**. A platform to estimate physical properties and thermal evolution pathways of exoplanet interiors. Abstract P51G-3440, Fall American Geophysical Union (AGU) Meeting 2019, San Francisco, CA. 
113. **H. Piet**<sup>PD</sup>, **H. Horn**<sup>GS</sup>, E. Greenberg, V. B. Prakapenka, P. R. Buseck, and **S.-H. Shim**<sup>PI</sup>. Challenges in high-pressure experiments on hydrogen in iron alloys-discrepancies among hydrogen sources. Abstract MR21A-07, Fall American Geophysical Union (AGU) Meeting 2019, San Francisco, CA.
112. **J. Dolinski**<sup>GS</sup>, **S.-H. Shim**<sup>PI</sup>, K. Leinenweber, and V. B. Prakapenka. Mineralogy of the silicon-rich mantle: implications for Mars and exoplanets. Abstract 3055, Lunar and Planetary Science Conference (LPSC) 2019, Houston TX.
111. **H. Piet**<sup>PD</sup>, **S.-H. Shim**<sup>PI</sup>, E. Greenberg, and V. B. Prakapenka. Hydrogen and sulfur in metallic iron at high pressure and high temperature and implications for the cores of Earth and Mars. Abstract DI43C-0031, Fall American Geophysical Union (AGU) Meeting 2018, Washington, DC. 
110. **J. O'Rourke**<sup>PD</sup>, C. Gillmann, P. J. Tackley, and **S.-H. Shim**. Chemistry Controls Dynamos in Metallic Cores: New Perspectives from Venus and Mars. Abstract GP12A-01, Fall American Geophysical Union (AGU) Meeting 2018, Washington, DC. 
109. S. Yu, E. Garnero, M. Li, and **S.-H. Shim**. Ultra-High Velocity Zones (UHVZs) at Earth's core mantle boundary. Abstract DI53A-0042, Fall American Geophysical Union (AGU) Meeting 2018, Washington, DC.
108. **S.-H. Shim**<sup>PI</sup>. Hydrogen in the deep interiors of Earth and exoplanets. Stanford Synchrotron Radiation Lightsource (SSRL) / Linac Coherent Light Source (LCLS) Users' Meeting and Workshops 2018, Menlo Park, CA. 
107. **S.-H. Shim**<sup>PI</sup>. New dense hydrous silica phases stable in the Earth's deep mantle. Synchrotron Infrared Spectroscopy on Materials in Extreme, National Synchrotron Light Source II (NSLS-II) and Center for Functional Nanomaterials (CFN) Users' Meeting 2018, Brookhaven, NY.
106. **S.-H. Shim**<sup>PI</sup>. Probing the valence state of Fe in mantle Mg-silicate perovskite with synchrotron X-ray. Advanced Photon Source (APS) / Center for Nanoscale Materials (CNM) Users Meeting 2018, Argonne, IL.
105. **B. Ko**<sup>GS</sup>, **S.-H. Shim**<sup>PI</sup>, V. B. Prakapenka, M. Kunz, and E. E. Alp. Incorporation of calcium into bridgmanite in Earth's lower mantle. Annual Meeting of Consortium for Materials Properties Research in Earth Sciences (COMPRES) 2018, Santa Ana Pueblo, NM.
104. **J. Dolinski**<sup>GS</sup>, **S.-H. Shim**<sup>PI</sup> and K. Leinenweber. Mineralogy of Martian interior with isotopic chemical model. Annual Meeting of Consortium for Materials Properties Research in Earth Sciences (COMPRES) 2018, Santa Ana Pueblo, NM. 
103. **B. Kulka**<sup>UG</sup>, **J. Dolinski**<sup>GS</sup>, **J. Tappan**<sup>UG</sup>, **S.-H. Shim**<sup>PI</sup>, and K. Leinenweber. The bridgmanite-majorite-akimotoite triple point measured in multi-anvil press and laser-heated diamond anvil cell. 

Annual Meeting of Consortium for Materials Properties Research in Earth Sciences (COMPRES) 2018, Santa Ana Pueblo, NM.

102. **J. G. O'Rourke**<sup>PD</sup> and **S.-H. Shim**. Suppressing the Martian dynamo with hydrogenation of the core by hydrated mantle minerals. Differentiation: Building the Internal Architecture of Planets 2018, Pasadena, CA. 
101. **J. G. O'Rourke**<sup>PD</sup> and **S.-H. Shim**. Suppressing the Martian dynamo with ongoing hydrogenation of the core by hydrated mantle minerals. Abstract 2390, 49th Lunar and Planetary Science Conference 2018, Houston, TX. 
100. **B. Ko**<sup>GS</sup>, **S.-H. Shim**<sup>PI</sup>, V. B. Prakapenka, M. Kunz, and E. E. Alp. Incorporation of calcium in bridgmanite in the deep mantle. Abstract MR43A-0479, Fall American Geophysical Union (AGU) Meeting 2017, New Orleans, LA.
99. **H. Piet**<sup>PD</sup>, **S.-H. Shim**<sup>PI</sup>, **J. Tappan**<sup>UG</sup>, K. Leinenweber, E. Greenberg, and V. B. Prakapenka. Dehydration of  $\delta$ -AlOOH in the lower mantle. Abstract MR43A-0462, Fall American Geophysical Union (AGU) Meeting 2017, New Orleans, LA.
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97. **C. Nisr**<sup>PD</sup>, **S.-H. Shim**<sup>PI</sup>, K. Leinenweber, and A. V. G. Chizmeshya. Raman spectroscopy of water-rich stishovite and dense high-pressure silica up to 55 GPa. Abstract MR43A-0457, Fall American Geophysical Union (AGU) Meeting 2017, New Orleans, LA.
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95. P. Piet, J. Badro, F. Nabiei, T. Dennenwaldt, **S.-H. Shim**, M. Cantoni, C. Hebert, and P. Gillet. Spin and valence dependence of iron partitioning in Earth's deep mantle. Abstract DI41C-2640, Fall American Geophysical Union (AGU) Meeting 2016, San Francisco, CA.
94. **S.-H. Shim**<sup>PI</sup>, **Y. Ye**<sup>PD</sup>, and V. B. Prakapenka. Inter-comparison of the gold, platinum, and MgO pressure scales up to 145 GPa and 2,500 K. Abstract MR12A-05, Fall American Geophysical Union (AGU) Meeting 2016, San Francisco, CA.
93. **S.-H. Shim**<sup>PI</sup>. Un-Earth-like interiors of Earth-like exoplanets. International Union of Crystallography (IUCr) High-Pressure Workshop, September 20–24, 2016, Pohang, Korea. 
92. **C. Nisr**<sup>PD</sup>, **S.-H. Shim**<sup>PI</sup>, K. Leinenweber, R. Hervig, V. B. Prakapenka, Y. Meng, and X. Liu. Stability of pure hydrous silica at geotherm temperatures up to 70 GPa. Abstract MR23B-2660, Fall American Geophysical Union (AGU) Meeting 2015, San Francisco, CA.
91. **H. Chen**<sup>GS</sup>, **S.-H. Shim**<sup>PI</sup>, K. Leinenweber, Y. Meng, and V. B. Prakapenka. Crystal structure of pure and aluminous calcium silicate perovskites at mantle related pressure and temperature. Abstract MR23B-2659, Fall American Geophysical Union (AGU) Meeting 2015, San Francisco, CA.
90. **B. Ko**<sup>GS</sup> and **S.-H. Shim**<sup>PI</sup>. Effects of compositional variation of basalt on subducting slabs over time. Abstract MR13C-2711, Fall American Geophysical Union (AGU) Meeting 2015, San Francisco, CA.

89. **S.-H. Shim**<sup>PI</sup>, **B. Grocholski**<sup>PD</sup>, **Y. Ye**<sup>PD</sup>, E. Alp, S. Xu, D. Morgan, Y. Meng, and V. B. Prakapenka. Low Fe<sup>3+</sup> in bridgmanite and possible existence of an oxidizing layer in the mid mantle. Abstract DI14A-04, Fall American Geophysical Union (AGU) Meeting 2015, San Francisco, CA.
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87. **S.-H. Shim**<sup>PI</sup>. Low Fe<sup>3+</sup> in bridgmanite in the mid mantle. Carbon at Extreme Conditions, Centre Européen de Calcul Atomique et Moléculaire (CECAM), 2015, Lugano, Switzerland.
86. **S.-H. Shim**<sup>PI</sup>, **C. Nisr**<sup>PD</sup>, **H. Chen**<sup>GS</sup>, **B. Ko**<sup>GS</sup>, M. D. Pagano, S. Desch, and P. A. Young. Un-Earth-like interiors of Earth-like exoplanets. Abstract #5020, Comparative Tectonics and Geodynamics of Venus, Earth, and Rocky Exoplanets, 2015, Pasadena, CA. 
85. A. Lorenzo, S. J. Desch, **S.-H. Shim**, and D. Nys. Effect of Fe redox state and Mg/Si ratio on exoplanet mass-radius relations. Abstract #2908, 46th Lunar and Planetary Science Conference 2015, Houston, TX. 
84. **C. Nisr**<sup>PD</sup>, **S.-H. Shim**<sup>PI</sup>, K. Leinenweber, R. L. Hervig, V. B. Prakapenka, and Y. Meng. Water in Al-free stishovite up to 65 GPa and 2000 K. Abstract MR21A-4302, Fall American Geophysical Union (AGU) Meeting 2014, San Francisco, CA.
83. **Y. Ye**<sup>PD</sup>, **C. Gu**<sup>GS</sup>, **S.-H. Shim**<sup>PI</sup>, V. B. Prakapenka, and Y. Meng. In situ measurements of the post-spinel and post-garnet phase boundaries in pyrolite at 17–32 GPa and 1500–2400 K. Abstract DI41B-4331, Fall American Geophysical Union (AGU) Meeting 2014, San Francisco, CA.
82. **H. Chen**<sup>GS</sup>, **S.-H. Shim**<sup>PI</sup>, K. Leinenweber, Y. Meng, and V. B. Prakapenka. Crystal structure of calcium silicate perovskite synthesized under water saturated conditions at mantle related pressure–temperature. Abstract MR21A-4301, Fall American Geophysical Union (AGU) Meeting 2014, San Francisco, CA.
81. **S.-H. Shim**<sup>PI</sup>. Un-Earth-like interiors of Earth-like exoplanets. Search for Life Beyond the Solar System. March 16–21, 2014, Tucson, AZ. 
80. **C. Nisr**<sup>PD</sup> and **S.-H. Shim**<sup>PI</sup>. Thermal expansion of SiC in the deep interiors of carbide exoplanets. Search for Life Beyond the Solar System. March 16–21, 2014, Tucson, AZ. 
79. **S.-H. Shim**<sup>PI</sup>, **Y. Ye**<sup>PD</sup>, V. B. Prakapenka, and Y. Meng. Effects of iron and aluminum on phase boundaries at 600–800 km depths. Abstract EGU2014-8674, European Geophysical Union Meeting 2014, Vienna, Austria.
78. **C. Nisr**<sup>PD</sup>, **S.-H. Shim**<sup>PI</sup>, K. Leinenweber, and V. B. Prakapenka. Effect of water on the compressional behaviors of SiO<sub>2</sub> stishovite up to 30 GPa. Abstract MR21A–2328, Fall American Geophysical Union (AGU) Meeting 2013, San Francisco, CA.
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72. **B. Grocholski**<sup>PD</sup>, **S.-H. Shim**<sup>PI</sup>, V. B. Prakapenka. Stability and compressibility of seifertite from 1 bar to 140 GPa. Abstract MR54A–08, Fall American Geophysical Union (AGU) Meeting 2012, San Francisco, CA.
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27. **S.-H. Shim**<sup>PI</sup>, R. Lamm, **S. Rekhi**<sup>PD</sup>, **K. Catalli**<sup>GS</sup>, **J. Santillán**<sup>PD</sup>, and **S. Lundin**<sup>GS</sup>. New micro-Raman spectroscopy systems for high-temperature studies in the diamond anvil cell. Abstract MR13B-02, Eos Transactions American Geophysical Union, 86, Fall Meeting Supplement, 2005, San Francisco, CA.
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## Media Highlights

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# Grants

**Tags:** Shim's recent developments are highlighted with the following symbols.

 Solar-system planets and exoplanets     Technical developments     Educational developments

- 2025/01–2026/12:** NASA-80NSSC25K7172, Sulfur in the Metallic Cores of Super-Earths and its Impact on their Magnetic Fields and Mass-Radius Relations. PI: S.-H. Shim. \$633,473; 50% recognition for Shim 
- 2024/08–2026/07:** NSF-AST2406790, NSF-BSF Application: Collaborative Research on Reaction between H<sub>2</sub>-H<sub>2</sub>O Fluid and Silicate, and Implications for Uranus and Neptune. PI: S.-H. Shim. \$350,984; 100% recognition for Shim 
- 2024/08–2026/07:** NSF-EAR2404972, EAGER: Vibrational Electron Energy Loss Spectroscopy - A New Nanometer Probe for Planetary Materials and Volatiles. PI: S.-H. Shim. \$293,045; 100% recognition for Shim 
- 2024/09–2027/08:** NASA-80NSSC25K7021, Iron and its Impact in Magma Oceans of Emerging Rocky Planets. PI: S.-H. Shim. \$524,788; 100% recognition for Shim  
- 2024/09–2027/08:** NSF- EAR2348824, Collaborative Research: Searching for the Origin of ULVZs beneath Antarctica. PI: S. Hansen. \$560,319; 10% recognition for Shim
- 2024/09–2026/08:** NSF- EAR2348824, Collaborative Research: Dynamic Compression of Iron-Sulfur Alloys at the Earth's Core Conditions. PI: S.-H. Shim. \$99,500; 50% recognition for Shim
- 2024/02/21–2025/02/20:** NSF- EAR2335071, EA: Upgrade of the Laser Heating System in the High-Pressure Diamond-Anvil Cell Laboratory at Arizona State University. PI: S.-H. Shim. \$63,872; 100% recognition for Shim 
- 2023/02/01–2026/12/31:** NASA-80NSSC23K0362, Hydrogen at the Core-Mantle Boundary of Mars. PI: S.-H. Shim. \$445,758; 100% recognition for Shim  
- 2022/10/01–2026/09/30:** NASA-80NSSC23K0265, Magma Oceans in Rocky Exoplanets: Understanding Their Dynamic and Magnetic Evolution with New Models and Experiments. PI: J. O'Rourke. \$542,369 
- 2022/08/15–2026/07/31:** NSF-EAR2153968, Collaborative Research: From Silicate Melts Properties to the Dynamics and Evolution of an Early Basal Magma Ocean. PI: W. Mao (Stanford U.). \$249,976 for Shim at ASU  
- 2022/01/11:** NSF-EAR2140416, Upgrade of the Raman Spectroscopy System at the High-Pressure Lab of Arizona State University. PI: S.-H. Shim. \$85,275; 100% recognition for Shim 
- 2021/10/01–2026/09/30:** NSF-EAR2131833, Facility for Open Research in a Compressed Environment (FORCE) at Arizona State University. PI: K. Leinenweber; co-PI: A. Navrotsky, T. Sharp, D. Smith, and S.-H. Shim. \$13,711,265; 5% recognition for Shim 
- 2021/08/01–2023/07/31:** NSF-AST2108129, Ingassing of Hydrogen in the Interiors of Sub-Neptunes and Gas Giants. PI: S.-H. Shim. \$320,615; 100% recognition for Shim 
- 2020/07/01–2022/06/30:** NSF-AST2005567, Effect of Hydrogen on the Sulfur-rich Martian Core. PI: S.-H. Shim. \$295,000; 100% recognition for Shim 
- 2020/07/01–2022/06/30:** NSF-EAR2019565, Possible Storage of H<sub>2</sub>O in Mantle Ca(Ti,Si)O<sub>3</sub> perovskite. PI: S.-H. Shim. \$279,262; 100% recognition for Shim
- 2019/08/01–2021/07/31:** NSF-EAR1921298, Effect of Hydrogen on the Behaviors of Fe Alloys in the Earth's Outer Core. PI: S.-H. Shim; co-PI: A. Chizmeshya. \$298,952; 80% recognition for Shim

**2019/06/01–2021/05/31:** NSF-EAR1855624, Cooperative Studies of The Earth's Deep Interior (CSEDI) Collaborative Research: Ultra-High Velocity Zones (UHVZs) at the Core-Mantle Boundary. PI: E. Garnero; co-PI: M. Li and S.-H. Shim. \$544,767; 33% recognition for Shim

**2018/01/03–2022/01/02:** NASA-80NSSC18K0353, Breaking the Barrier between Ice and Rock – New Mass-Radius Relations for Gas Giant, Icy Giant, Mini-Neptune, and Water World Exoplanets. PI: S.-H. Shim; co-PI: K. Leinenweber and A. Chizmeshya. \$559,951; 40% recognition for Shim

**2014/12/31–2020/12/30:** NASA, Exoplanetary Ecosystems: Exploring Life's Detectability on Chemically Diverse Exoplanets. PI: S. Desch; co-PI: S.-H. Shim among others. \$6,097,436; 4% recognition for Shim

**2017/06/15–2020/06/15:** NSF-EAR1725094, Calcium in Bridgmanite in the Deep Mantle. PI: S.-H. Shim. \$271,735; 100% recognition for Shim

**2017:** Stanford Linear Accelerator Center (SLAC) National Accelerator Laboratory, Materials Synthesis for Laser-Driven Shock Wave Experiments at Linac Coherent Light Source (LCLS). PI: S.-H. Shim. \$22,248; 100% recognition for Shim

**2016/01/01–2019/12/31:** The Keck Foundation, Water from the Heavens: The Origins of Earth's Hydrogen. PI: P. Buseck; co-PI: S.-H. Shim among others. \$1,500,000; 14% recognition for Shim

**2014/09/01–2016/08/31:** NSF-EAR1401270, Cooperative Studies of The Earth's Deep Interior (CSEDI) Collaborative Research: Deep Mantle Cycling of Oceanic Crust. PI: E. Garnero; co-PI: A. McNamara and S.-H. Shim. \$818,541; 33% recognition for Shim

**2013/09/01–2018/08/31:** NSF-FESD-Type I, The Dynamics of Earth System Oxygenation. PI: A. Anbar; co-PI: S.-H. Shim among others. \$3,080,000; 5% recognition for Shim

**2013/08/01–2016/07/31:** NSF-EAR1321976, Hydration of Dense Polymorphs of Silica in Subducting Slabs. PI: K. Leinenweber; co-PI: Shim. \$312,839; 33% recognition for Shim

**2011/01/01–2015/12/31:** NSF-EAR1301813, The Perovskite to Post-Perovskite Phase Boundary in Mantle Rocks. PI: S.-H. Shim. \$398,149 (\$229,068 transferred to ASU); 100% recognition for Shim

**2010/09/01–2015/08/31:** NSF-EAR1316022, Cooperative Studies of The Earth's Deep Interior (CSEDI) Collaborative Research: Valence State of Iron in the Lower Mantle. PI: S.-H. Shim; co-PI: Morgan. \$330,922 for Shim (\$69,447 transferred to ASU)

**2009/12/15–2014/11/30:** NSF-EAR1316007, Understanding the Complexity near the 660-km Seismic Discontinuity. PI: S.-H. Shim. \$360,281 (\$58,468 transferred to ASU); 100% recognition for Shim

**2008/07/01–2011/06/30:** NSF-EAR0757871, Cooperative Studies of The Earth's Deep Interior (CSEDI) Collaborative Research: Multi-scale Analysis of Mantle Discontinuities using Inverse Scattering of SS Waves and Experimental Mineral Physics. PI: R. van der Hilst; co-PI: S.-H. Shim. \$299,999

**2010/06/01–2011/05/30:** NSF-DMR0819762, CMSE-Initiative 3. S.-H. Shim as a co-PI. \$26,500 for Shim

**2008/01/01–2010/12/31:** NSF-EAR0738655, Equation of State and Phase Boundary of Post-Perovskite. PI: S.-H. Shim. \$259,803; 100% recognition for Shim

**2004/07/01–2008/06/30:** NSF-EAR0337005, In situ Raman Spectroscopy Study for Phase Diagrams of Mantle Minerals at High Pressure and Temperature. PI: S.-H. Shim. \$255,829; 100% recognition for Shim

**2004/03/10–2006/03/09:** NSF-EAR0337156, Acquisition of a Combined Micro-Raman Spectroscopy and Laser Heating System for in situ High Pressure and High Temperature Studies. PI: S.-H. Shim. \$185,244; 100% recognition for Shim

**2004:** Wade Award, Acquisition of a Double Monochromator for Raman Spectroscopy Measurements at

High Pressure and Temperature. PI: S.-H. Shim. \$45,000; 100% recognition for Shim

# Courses

**Tags:** Shim's recent developments are highlighted with the following symbols.

 Solar-system planets and exoplanets     Technical developments     Educational developments

All the courses here are in-person. F: Fall, S: Spring

|                                                                                                                         |                                                                                                                                                                         |
|-------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>2026S, 2024F, 2023F, 2022F, 2021F, 2020F:</b> SES230. Coding for Exploration, ASU                                    |   |
| <b>2025F:</b> SES598. SESE Knowledge Exploration, ASU                                                                   |                                                                                      |
| <b>2025F, 2024S, 2022S, 2018F, 2016F, 2014F:</b> GLG494/598. Advanced Mineralogy: Crystallography and Spectroscopy, ASU |                                                                                                                                                                         |
| <b>2025S, 2023S:</b> SES494/598. Planetary Interiors, ASU                                                               |   |
| <b>2021S:</b> SES494/598. Data and Computation, ASU                                                                     |   |
| <b>2020S, 2017F, 2015F:</b> GLG494/598. Introduction to Mineral Physics, ASU                                            |                                                                                      |
| <b>2019F:</b> SES130. Coding for Exploration, ASU                                                                       |                                                                                      |
| <b>2018S:</b> GLG494/598. Exploring Data with Python, ASU                                                               |                                                                                      |
| <b>2017S:</b> GLG294/598. Python for Earth Science, ASU                                                                 |                                                                                      |
| <b>2016S, 2013S:</b> GLG101. Introduction to Geology I (Physical), ASU                                                  |                                                                                                                                                                         |
| <b>2015S, 2014S:</b> GLG305. Dynamic Earth, ASU (co-teach with D. DeVecchio in 2014S)                                   |                                                                                                                                                                         |
| <b>2014F:</b> GLG591. Archean Geophysics (co-teach with A. Anbar), ASU                                                  |                                                                                                                                                                         |
| <b>2014S:</b> GLG591. Chemical Processes in Earth's Interior, ASU                                                       |                                                                                                                                                                         |
| <b>2013F:</b> GLG598. Planetary Materials, ASU                                                                          |                                                                                                                                                                         |
| <b>2011S:</b> 12.591. Hydrogen: from Planetary to Energy Sciences, MIT                                                  |                                                                                    |
| <b>2010F:</b> 12.591. Volatiles in the Earth and Planetary Interiors, MIT                                               |                                                                                    |
| <b>2010S–2007S:</b> 12.108. Structure of Earth Materials, MIT                                                           |                                                                                                                                                                         |
| <b>2009F:</b> 12.591. The Core-Mantle Boundary, MIT                                                                     |                                                                                                                                                                         |
| <b>2008F, 2004F:</b> 12.575. Introduction to Mineral Physics, MIT                                                       |                                                                                                                                                                         |
| <b>2007S, 2006F, 2006S:</b> 12.080. EAPS Undergraduate Seminar, MIT                                                     |                                                                                                                                                                         |
| <b>2005F:</b> 12.571. Deep Water - Geophysical Prospective, MIT                                                         |                                                                                                                                                                         |
| <b>2005S:</b> 12.581. Phase Transitions in the Earth's Interior, MIT                                                    |                                                                                                                                                                         |
| <b>2004S:</b> 12.570. Structure and Dynamics of the Core-Mantle Boundary Region, MIT                                    |                                                                                                                                                                         |

# Mentoring

## B.S. Thesis

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**2027:** Aidan Ortega, Barrett, ASU (expected for May 2027)

**2025:** Ava Campbell, Barrett, ASU

**2023:** Shradha Ravikumar, Barrett, ASU

**2019:** Britany Kulka, ASU (now Ph.D. student at Oxford University)

**2015:** Shaela Noble, Barrett Honors College, ASU (now M.S. student at the University of California, Davis, Bidstrup Undergraduate Fellow)

**2007:** Caitlin Murphy, MIT (now Senior Policy Analyst at the National Renewable Energy Laboratory in Golden, CO)

## M.S. Thesis

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**2019–2021:** Britany Kulka, ASU (now postdoctoral researcher at Los Alamos National Laboratory)

**2018–2019:** Jonathan Dolinschi, ASU (now postdoctoral researcher at Los Alamos National Laboratory)

## Ph.D. Thesis

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**2025–present:** Alem Tukić, ASU (expected for 2030)

**2021–2025:** Xuehui Wei, ASU

**2018–2022:** Harrison Horn, ASU

**2017–2021:** Taehyun Kim, Yonsei University

**2014–2020:** Byeongkwan Ko, ASU (now postdoctoral researcher at Michigan State University)

**2014–2019:** Huawei Chen, ASU (now Assistant Professor at China University of Geosciences, China)

**2005–2011:** Krystle Catalli, MIT (now Principal Engineer at Apple, Inc)

## Postdoctoral Researchers

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**2025–present:** Allison Pease, ASU

**2022–present:** Taehyun Kim, ASU

**2026:** Xuehui Wei, ASU

**2023–2025:** Sibon Chen, ASU

**2020–2022:** Suyu Fu, ASU (now Japan Society for the Promotion of Science (JSPS) postdoctoral fellow at the University of Tokyo)

**2017–2021:** Hélène Piet, ASU (now research scientist at W. L. Gore & Associates)

**2020–2021:** Byeongkwan Ko, ASU

**2017–2019:** Joseph O'Rourke, Exploration Postdoctoral Fellow, ASU (co-advised with Prof. L. Elkins-Tanton; now Associate Professor at ASU)

**2016–2018:** Cayman Unterborn, Exploration Postdoctoral Fellow, ASU (co-advised with Prof. S. Desch; now research associate at ASU)

**2012–2018:** Carole Nisr, ASU (now Professor at Phoenix College)

**2012–2014:** Yu Ye, ASU (now Associate Professor at China University of Geosciences, China)

**2011:** Antonio Buono, MIT (now research scientist at ExxonMobil)

**2008–2011:** Brent Grocholski, MIT (now Senior Editor in journal *Science*)

**2006–2007:** Justin Hustoft, MIT (now Assistant Professor at Mount Mary University, WI)

**2004–2006:** Javier Santillán, MIT, Ford Postdoctoral Fellow (now Global CT Manager at Apple, Inc)

**2003–2005:** Sandeep Rekhi, MIT (now Principal Engineer at Apple, Inc)

## **Undergraduate Research Program**

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**I advised all the following undergraduate students for the undergraduate research program in my group.**

**2026–present:** Aidan Ortega, Barrett, ASU

**2026–present:** Kassandra Amezcua, Barrett, ASU

**2024–2026:** Ava Campbell, Barrett, ASU (now Ph.D. student at Stanford)

**2024–2025:** Hamza Ayaz, ASU

**2023–2024:** Erin Mann, ASU

**2021–2023:** Shradha Ravikumar, Barrett, ASU

**2022:** Enzo Carrascal, ASU (now Ph.D. student at Caltech)

**2019:** Robert Rezvani and Ayla Zustra, Barrett, ASU

**2018–2019:** Britany Kulka, ASU (now Ph.D. student at Oxford University)

**2017–2019:** Jacqueline Tappan, ASU (now Staff Hydrogeologist at Leonard Rice Engineers Inc.)

**2017:** Jonathan Dolinschi, ASU (now Ph.D. student at Bayreuth University, Germany)

**2016–2017:** Abigail Weibel, ASU (now Project Manager Associate at ASU)

**2015–2016:** Patrick Kennedy, ASU (now Technologist at Audiolex)

**2015–2016:** Shaela Noble, ASU (now M.S. student at the University of California, Davis)

**2015–2016:** Mark Williamson, ASU

**2009:** Elizabeth George, MIT (now Detector Engineer at European Southern Observatory, Munich, Germany)

**2008:** Michael DeMeo, Leslie C. Patron Undergraduate Research Opportunity Program, MIT (now Application Engineer at Exa Corporation, Detroit, MI)

**2008:** Deidre LaBounty, MIT (now graduate student at the University of Alaska Fairbanks)

**2007–2008:** Rachel Zucker, Leslie C. Patron Undergraduate Research Opportunity Program, MIT (now senior research scientist at Kernel, CA)

**2006:** Sarah Slotznick, Leslie C. Patron Undergraduate Research Opportunity Program, MIT (now Miller postdoctoral fellow, Department of Earth and Planetary Science, University of California, Berkeley)

**2004–2005:** Nicholas Leiby, MIT (now Lead Data Scientist at Kyruus, Alexandria, VA)

## Visiting Graduate Students

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**2017–2021:** Taehyun Kim, Visiting Ph.D. student from Yonsei University, Korea

**2012–2013:** Qian Zhang, Visiting Ph.D. student from Peking University, China (now staff scientist at China University of Geosciences)

## Thesis Committee Member

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These students are not directly mentored by Shim. Shim was a committee member for the dissertation.

**2025:** Stephanie Sparks (Ph.D.), ASU

**2025:** Kevin Trinh (Ph.D.), ASU

**2022:** Qian Yuan (Ph.D.), ASU

**2021:** Jake Hanson (Ph.D.), ASU

**2020:** Hannah Shamloo (Ph.D.), ASU

**2019:** Hongyu Lai (Ph.D.), Shule Yu (Ph.D.), Alyssa Anderson (Ph.D.), Sarah Dillon (M.S.), Emily Garhart (M.S.), and Gabriel Franco (M.S.), ASU

**2018:** Alejandro Lorenzo (M.S.), ASU

**2016:** Shkolyar Svetlana (Ph.D.), ASU

**2015:** Jeffrey Lockridge (Ph.D.) Rebecca Smith (Ph.D.), and Alex Mastrean (B.S.), ASU

**2014:** Mingming Li (Ph.D.) and Michael Pagano (Ph.D.), ASU

**2011:** Michael Krawczynski (Ph.D.), and Qin Cao (Ph.D.), and Stephanie Brown (M.S.), MIT

**2010:** Jay Barr (Ph.D.), MIT

**2007:** Emily Van Ark (Ph.D.), Guangping Xu (Ph.D.), and Ping Wang (Ph.D.), MIT

**2006:** Shichun Huang (Ph.D.), MIT

## Graduate Students for Second Comprehensive Exam Project

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SESE at ASU (and also MIT) requires Ph.D. students to carry out with two separate research projects with two different faculty advisors for their comprehensive exams. Their second projects should involve work in substantially different fields. Shim has regular weekly research meetings with the students for at least 1 semester to ensure sufficient progress in their second research projects.

**2025–2026:** Guangpu Yi

**2023–2024:** Prachi Kar

**2022–2023:** Leah Shteynman

**2018–2019:** Emily Garhart and Gabriel Franco, ASU

**2017–2018:** Jonathan Hoh and Alexandra Pye, ASU

**2015–2017:** Alyssa Anderson, ASU

**2008–2012:** Xuefeng Shang and Nathaniel Dixon, MIT

**2004–2006:** Rosalee Lamm, MIT

## Comprehensive Examination Committees

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**2024–2025:** Marco Lalonde, ASU

**2023–2024:** Prachi Kar and Yoav Rotman, ASU

**2021–2022:** Sasha Sypkens and Kevin Trinh, ASU

**2020–2021:** Camerian Millsaps, Saira Hamid, and Aditya Khuller, ASU

**2019–2020:** Mariah Heck, Qian Yuan, and Harrison Horn, ASU

**2018–2019:** Alexandra Pye, Jonathan Hoh, and Sierra Ferguson, ASU

**2017–2018:** Hanna Shamloo and Jake Hanson (Chair), ASU

**2016–2017:** Huawei Chen, Byeongkwan Ko, Kara Brugman, and Alyssa Anderson, ASU

**2015–2016:** Megan Miller (Chair), Julie Mitchell, and Hongyu Lai, ASU

**2014–2015:** Luke Probst, Ruirui Han, Jinping Hu (Chair), and Christopher Haberle, ASU

**2013–2014:** Jeffrey Lockridge and Jinping Hu, ASU

**2012:** Chen Gu, MIT

**2009:** Qin Cao (Chair), Nathan Dixon, and Xuefeng Shang, MIT

**2007:** Krystle Catalli, MIT

**2006:** Jay Barr and Noah McLean, MIT

# Service

**Tags:** Shim's recent developments are highlighted with the following symbols.

 Solar-system planets and exoplanets     Technical developments     Educational developments

## University Service

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**2022–2023:** Member of Faculty Search Committee, Facility for Open Research in a Compressed Environment (FORCE), ASU

**2022:** Member of Research Scientist Search Committee, Facility for Open Research in a Compressed Environment (FORCE), ASU

**2021–present:** Member of Steering Committee, Facility for Open Research in a Compressed Environment (FORCE), ASU 

**2019–present:** Member of Steering Committee, Center for Materials of the Universe (MOTU), ASU 

**2019:** Chair of Organizing Committee for Workshop on Materials of the Universe (MOTU), ASU 

**2017–2022:** Member of Materials Governance Board, ASU

## Unit/Department Service at ASU

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**2025–present:** Chair of Graduate Student Committee, SESE, ASU

**2024–2025:** Member of Graduate Student Committee, SESE, ASU

**2021–2024:** Member of SESE Annual Evaluation Committee, SESE, ASU

**2023–2024:** Member of SESE Award Committee, SESE, ASU

**2020–2023:** Member of SESE Exploration Postdoctoral Scholar Committee, SESE, ASU

**2019–2020:** Member of Thermochemistry Faculty Search Committee, School of Molecular Sciences (SMS), ASU

**2019–2020:** Member of Faculty Annual Evaluation Committee, SESE, ASU

**2016–2017:** Member of Geophysics Faculty Search Committee, SESE, ASU

**2016–2017:** Member of Geochemistry Faculty Search Committee, SESE, ASU

**2016:** Chair of Graduate Student Recruiting Committee, SESE, ASU

**2013–2016:** Member of Graduate Student Recruiting Committee, SESE, ASU

**2012–2013:** Chair of Geophysics Faculty Search Committee, SESE, ASU

**2012:** Member of Award Committee, SESE, ASU

## Unit/Department Service at MIT

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**2009–2010:** Member of Graduate Admissions Committee, EAPS, MIT

**2008–2009:** Member of Graduate Committee, EAPS, MIT

**2006–2007:** Chair of Independent Activities Period Educational Program Committee, EAPS, MIT

**2007:** Member of Promotion Committee for Chatterjee to Senior Research Scientist, EAPS, MIT

**2006:** Member of Theoretical Geophysics Faculty Search Committee, EAPS, MIT

**2003–2006:** Member of Graduate Admissions Committee, EAPS, MIT

## Non-ASU Professional Service

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**2024–present:** Committee Member, American Geophysical Union Mineral and Rock Physics Canvassing Committee 

**2024–present:** Committee Member, Synchrotron Earth and Environmental Science User Community 

**2022:** Organizer, Town Hall Meeting for Facility for Open Research in a Compressed Environment (FORCE), 2022 

**2019:** Organizer, ED52A - Linking Education and Research with Jupyter, Fall American Geophysical Union Meeting, San Francisco, December 9–13, 2019 

**2019:** External Reviewer for Geophysics Faculty Search, Yonsei University, Korea

**2018–present:** Member, NExSS (Nexus for Exoplanet System Science) Steering Committee, NASA 

**2017:** Organizing Committee Chair, COMPRES Workshop: Software Toolkit, Consortium for Materials Properties Research in Earth Sciences (COMPRES) Annual Meeting, Hyatt Regency Tamaya Resort, Santa Ana Pueblo, NM, July 9–12, 2017 

**2017:** Member, Review panel for the Cooperative Studies of the Earth's Deep Interior (CSEDI) program, National Science Foundation (NSF)

**2015:** Organizing Committee Chair, Workshop for US Large Multi-Anvil Press Facility (LMAPF), Consortium for Materials Properties Research in Earth Sciences (COMPRES) Annual Meeting, Colorado Springs, CO, July 6, 2015 

**2015–2018:** Member, Facilities Committee, Consortium for Materials Properties Research in Earth Sciences (COMPRES)

**2013–2015:** Editor, Geophysical Journal International

**2013–2020:** Member, Advanced Photon Source Proposal Review Panel for High Pressure Science, Argonne National Laboratory

**2010–2015:** Member, Executive Committee, Mineral and Rock Physics Focus Group, American Geophysical Union (AGU)

**2010–2014:** Vice Chair, Subcommittee for Spectroscopy, Diffraction, and New Instrumentations in Mineral Physics, Commission of Physics of Minerals, International Mineralogical Association (IMA)

**2010:** Consulting editor, Odyssey - Adventures in Science, Science Magazine for Middle Schoolers, “That Rocks!” issue

**2004–2010:** Member, Infrastructure Development Committee, Consortium for Materials Properties Research in Earth Sciences (COMPRES)

**2008:** Member, International Organizing Committee, 4th Asian Conference in High Pressure Research, Seoul, Korea

**2004–2006:** Associate Editor, American Mineralogist

## Journal Article Review

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Nature, Science, Nature Geoscience, Science Advances, Geophysical Research Letters, Proceedings of

the National Academy of Sciences, Journal of Geophysical Research, American Mineralogist, Earth and Planetary Science Letters, Physics of the Earth and Planetary Interiors, Physical Review, Journal of Solid State Chemistry

## **Grant Proposal Review**

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National Aeronautics and Space Administration (NASA) Exoplanet Research Program, National Science Foundation (NSF) Geophysics, National Science Foundation (NSF) Geochemistry and Petrology, National Science Foundation (NSF) Earth Sciences: Instrumentation and Facilities, National Science Foundation (NSF) Materials Research, European Science Foundation EuroMinSc